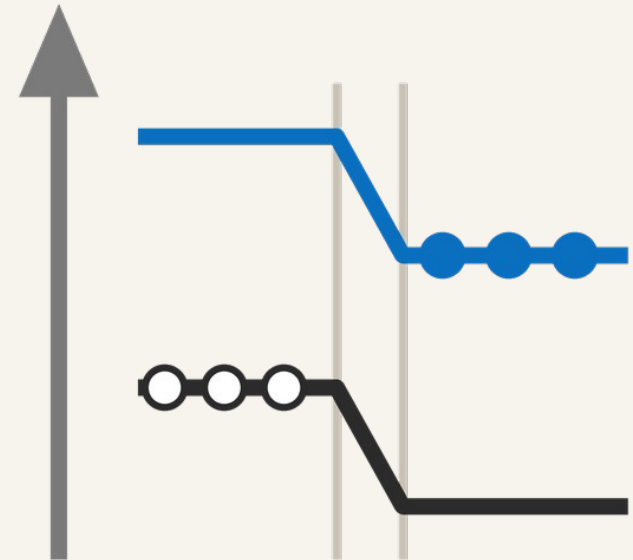


Semiconductors & p-n Junctions

Why some solids conduct, some don't, and some can be talked into it.

Higher Physics



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

2009 Higher Physics: scan the code and answer these.



Q8

Q11

Section A · charged particles,
capacitors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

PART 1

Energy bands

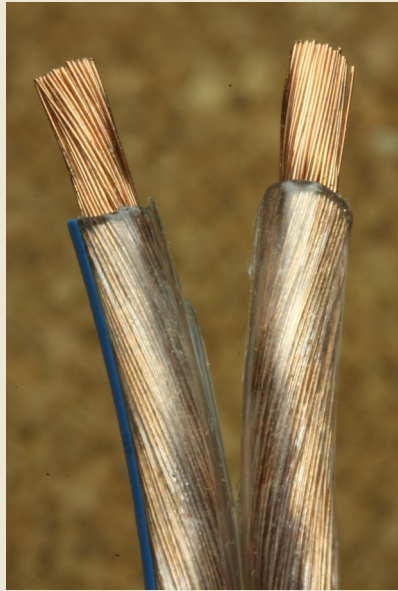
Why does copper conduct, glass not, and silicon only a little?

By the end of this part...



- 1 I can describe how energy bands form from the energy levels of single atoms.
- 2 I can name and describe the conduction band and the valence band.
- 3 I can use band structure to explain why a solid conducts or does not.

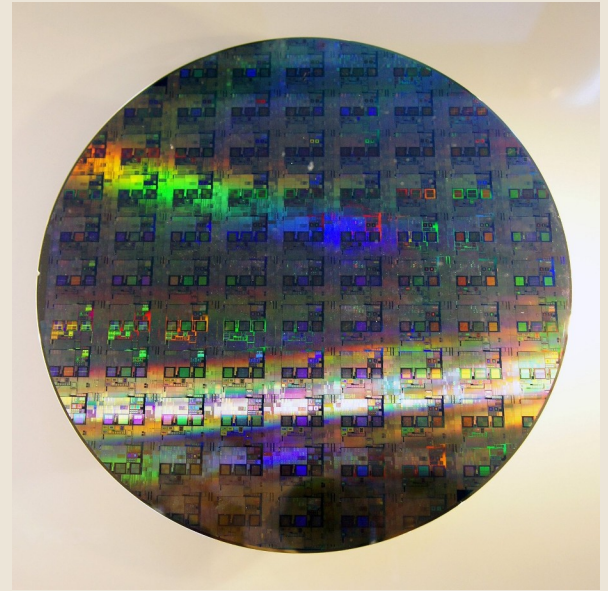
Which of these conducts best, and which worst?



copper wire



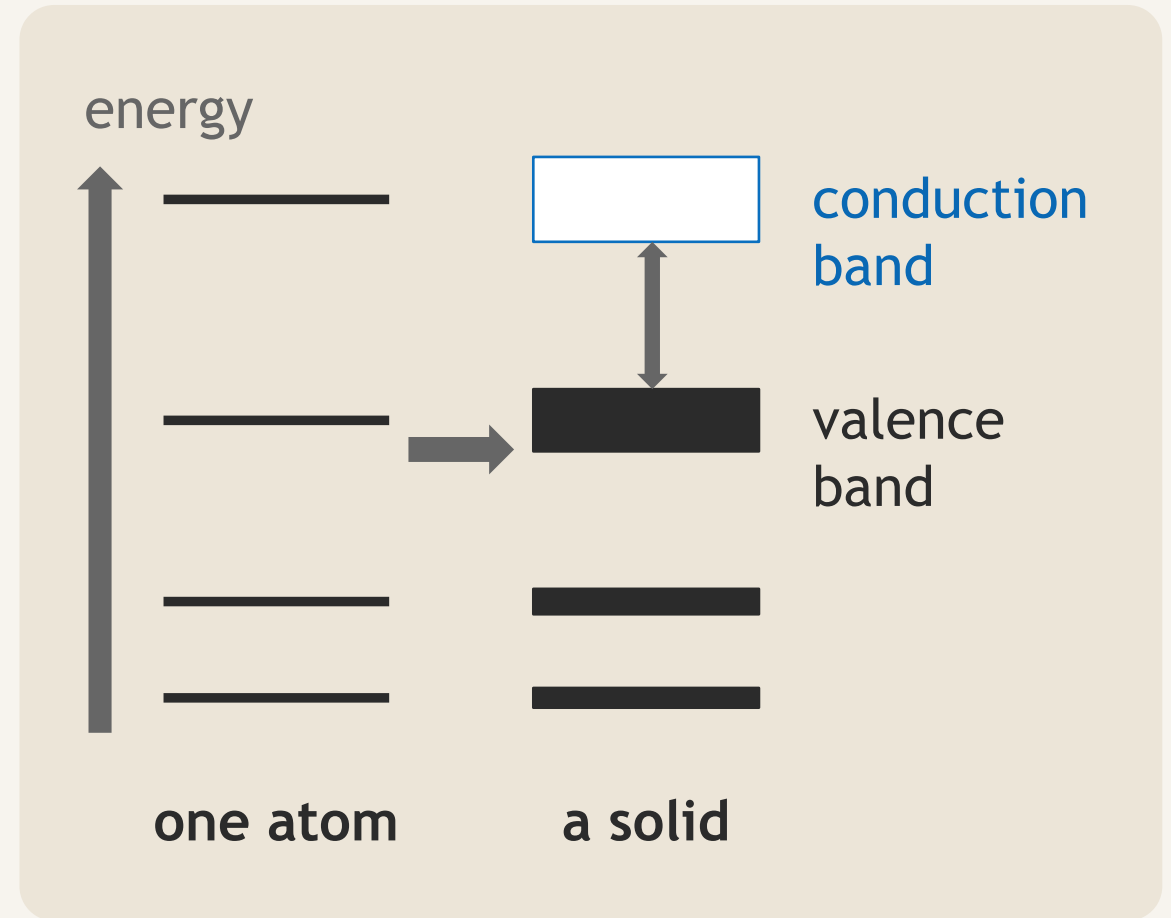
glass insulators



a silicon wafer

From energy levels to energy bands

- In a single atom, electrons sit in fixed energy levels.
- A solid is millions of atoms: the levels merge into bands.
- Between the bands are gaps no electron may have.

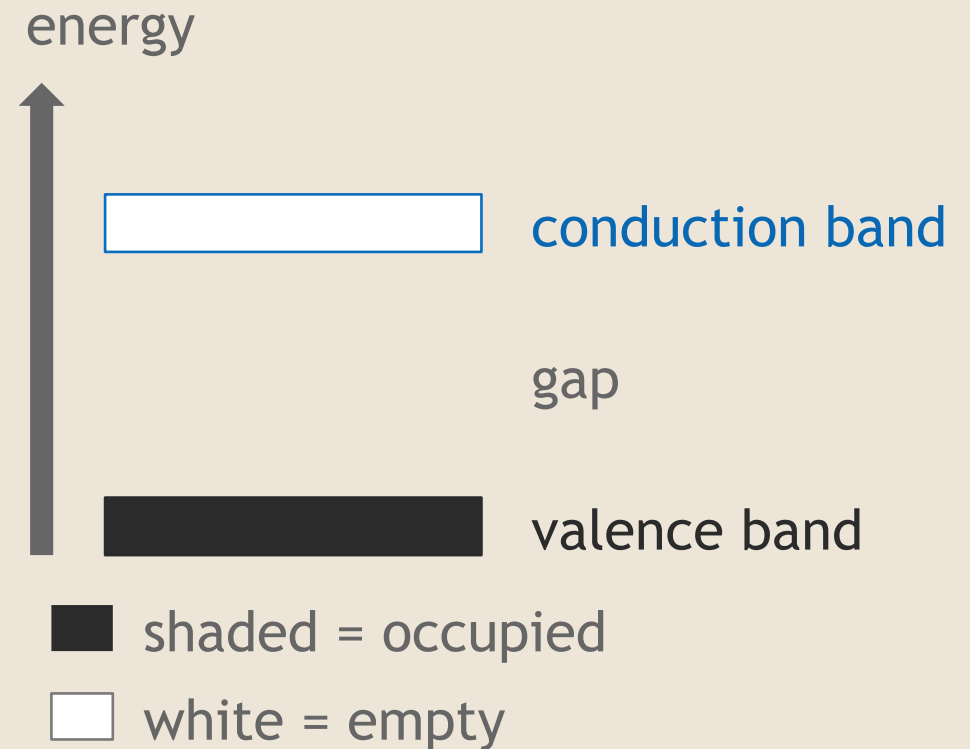


The conduction band and the valence band



The valence band holds the electrons that bond the solid together.

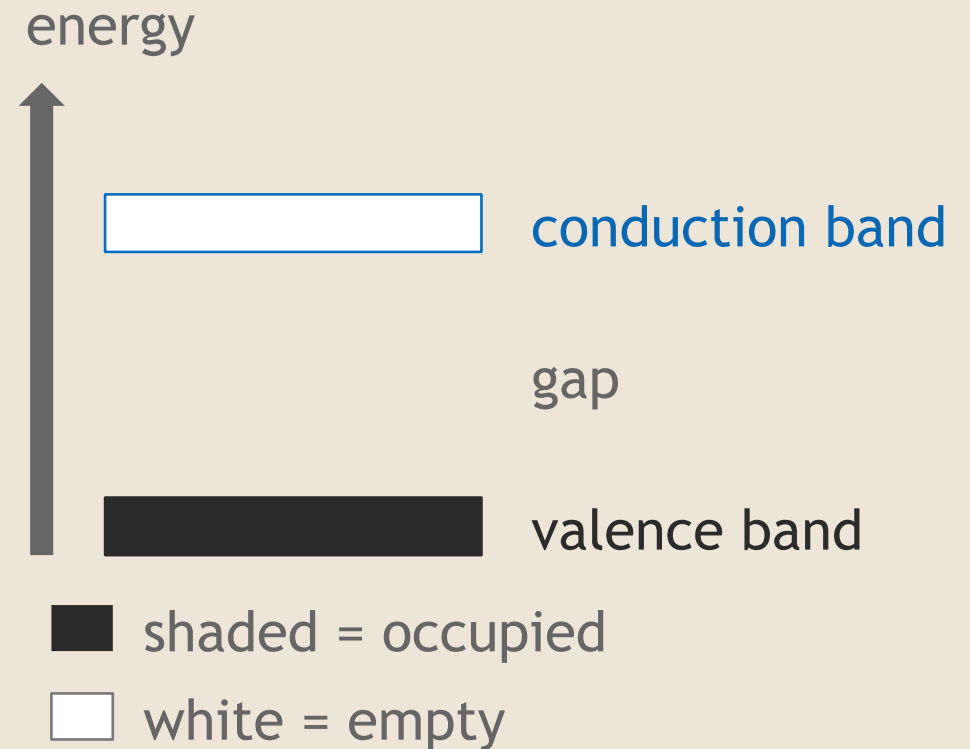
The conduction band is the band an electron has to reach before it can move freely and carry a current.



The gap between the bands



Between them is a gap – a range of energies no electron is allowed to have.

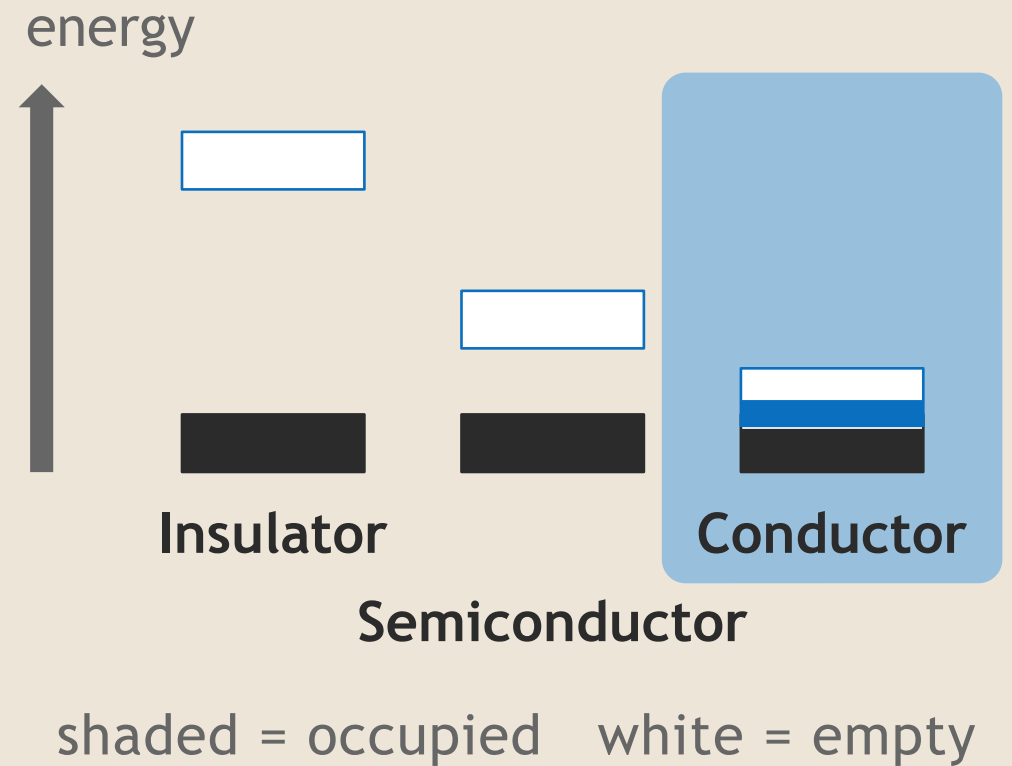


Conductors



In a conductor the conduction band and the valence band overlap, or the highest occupied band is only partly full.

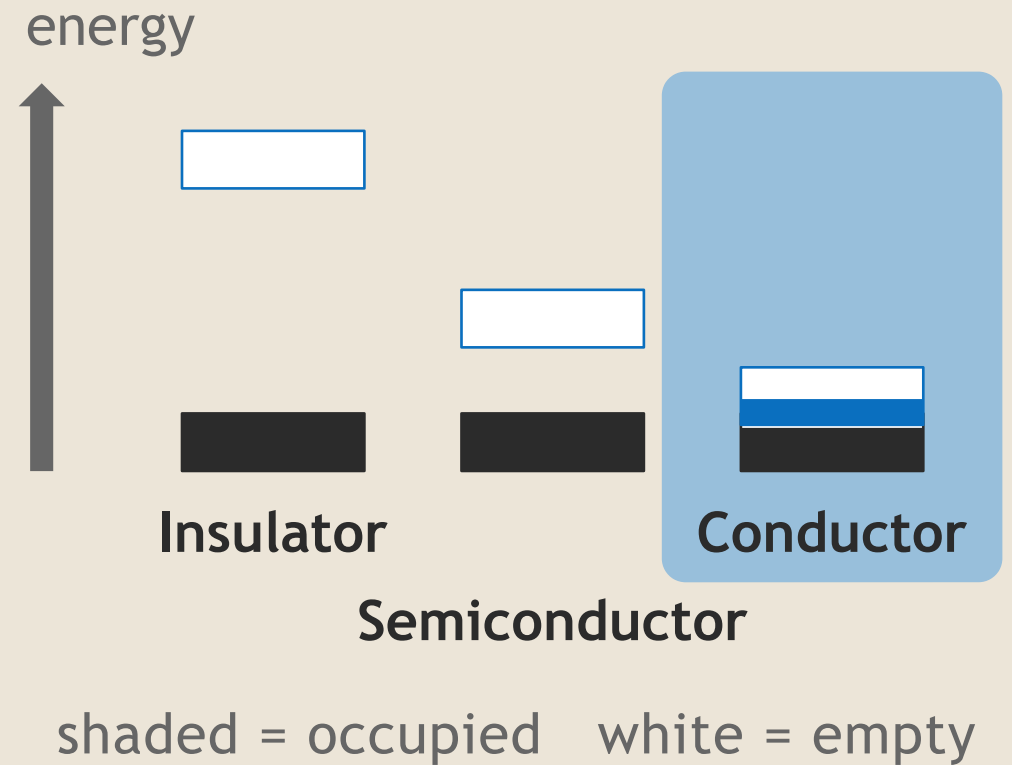
There are electrons free to move AND empty states for them to move into.



Why a conductor conducts well



There is no gap to cross, so a conductor conducts very well.



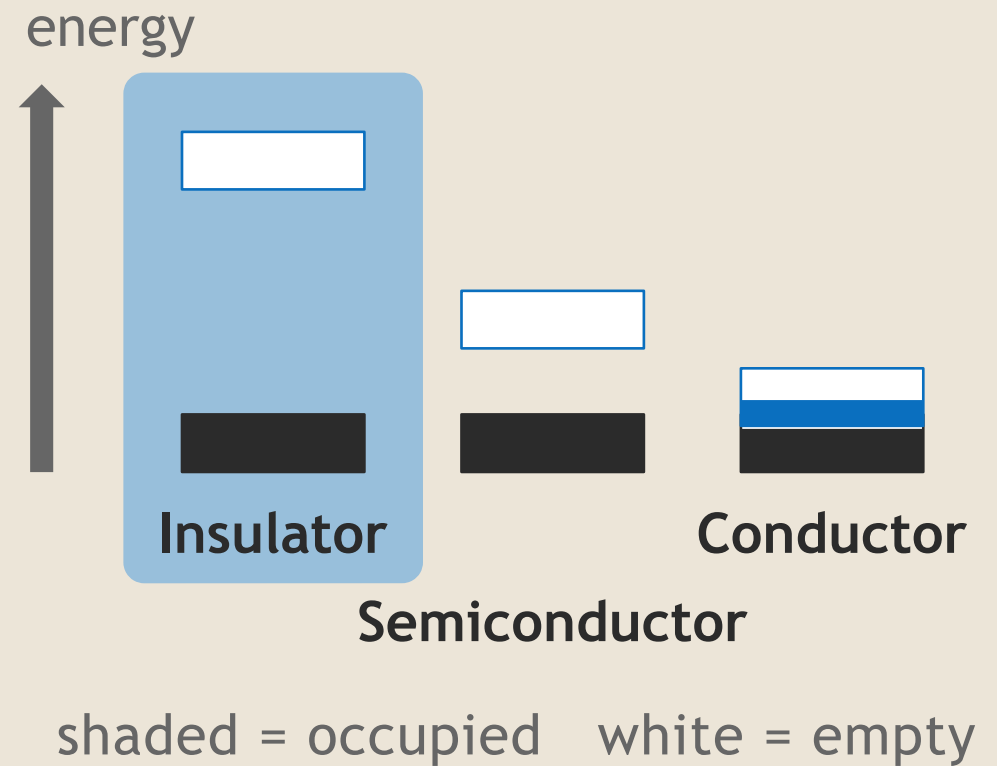
Insulators



In an insulator the valence band is full and the gap above it is very large.

At room temperature no electron has enough energy to cross that gap.

The conduction band stays empty, so no current flows.

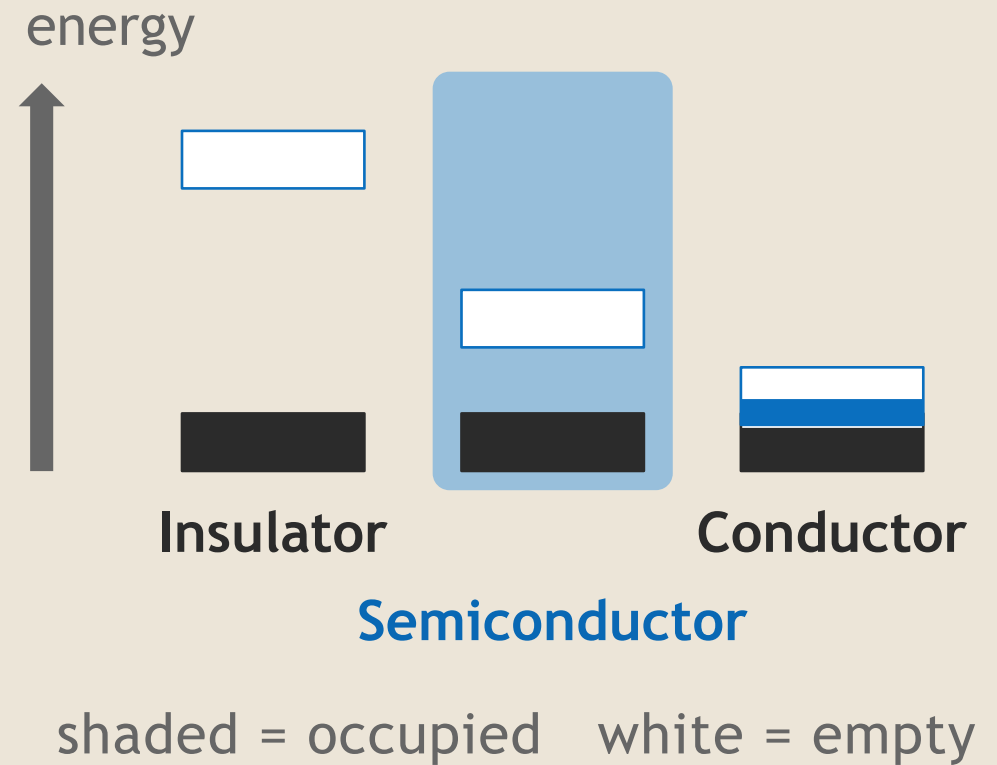


Semiconductors



In a semiconductor the valence band is full, but the gap above it is small.

At room temperature a few electrons have enough energy to cross it.

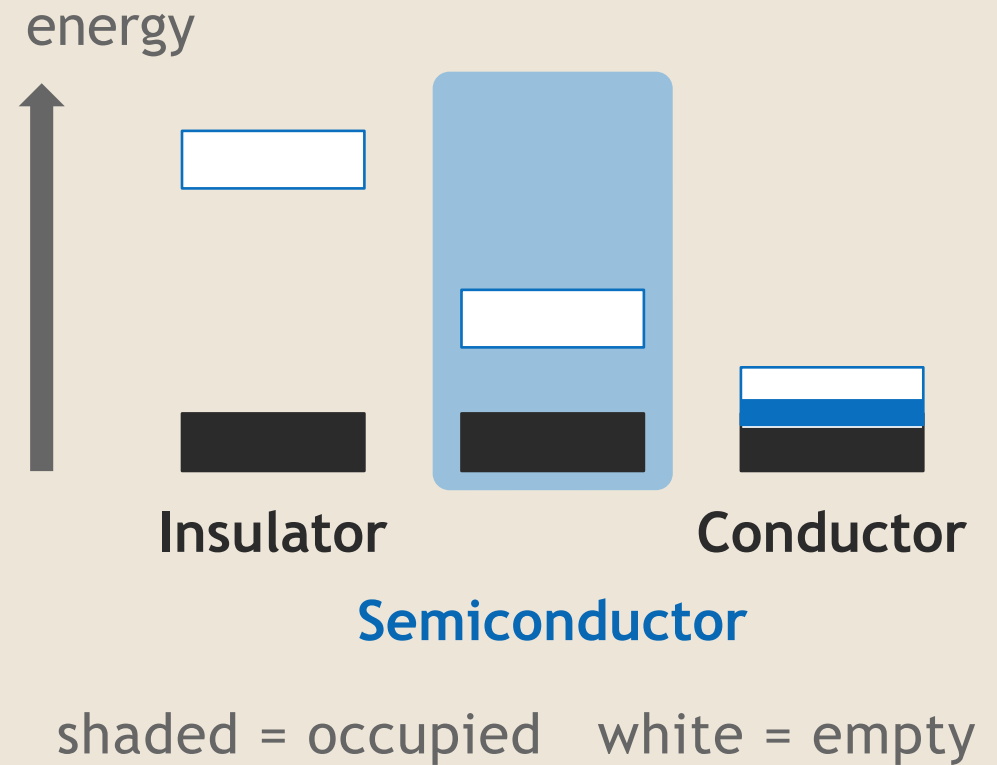


How well a semiconductor conducts

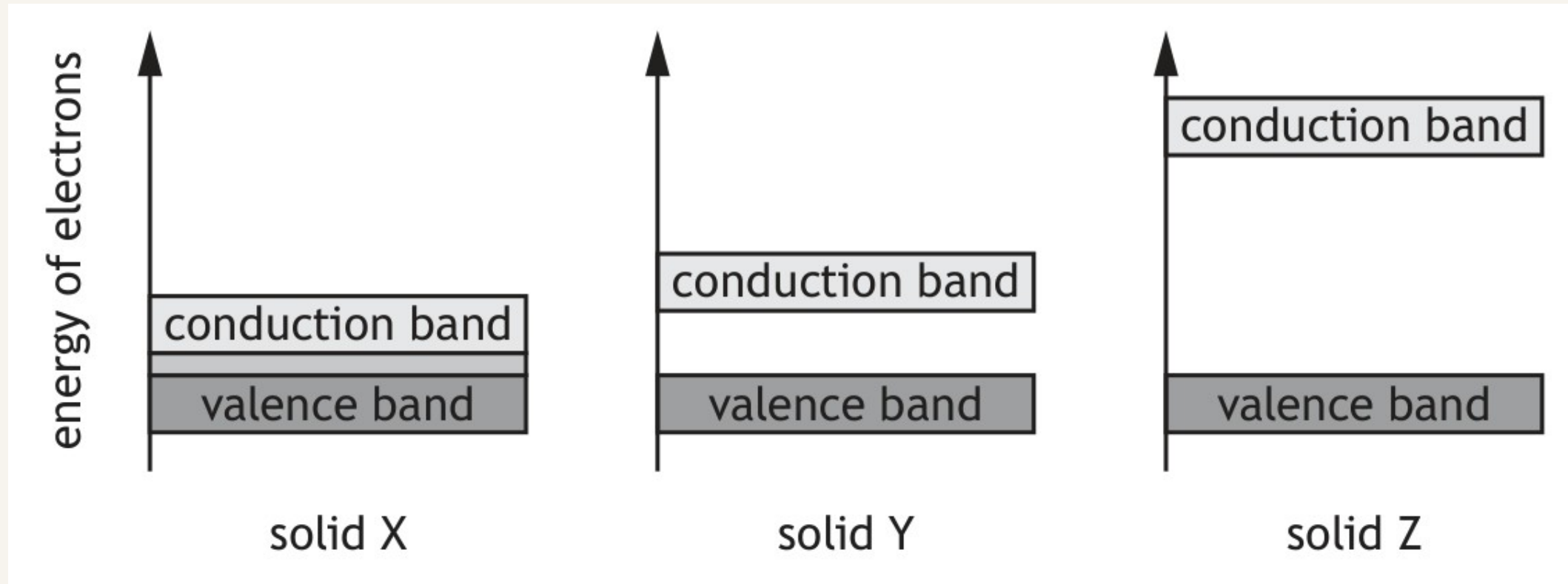


That gives a little conduction — more than an insulator, far less than a metal.

Warm it up and more electrons cross the gap, so a semiconductor conducts better when it is hotter.



The exam diagram: bands named, not shaded



In a semiconductor, the energy gap between the valence band and the conduction band is



- A** small, allowing some conduction to take place at room temperature
- B** large, allowing some conduction to take place at room temperature
- C** zero, allowing electrons to move freely
- D** large, meaning that no conduction can take place at room temperature
- E** small, meaning that no conduction can take place at room temperature.

Show Me: classify the solid. Answer on your board, one at a time.



1

State which band an electron must reach before it can carry a current.

2

A solid has a full valence band and a very large gap. State what kind of solid it is, and why it does not conduct.

3

State the two things a solid needs before it can conduct.

Show Me: classify the solid. Answer on your board, one at a time.



- 4 Explain why a semiconductor conducts better when it is warmed up.
- 5 A solid's conduction band and valence band overlap. State what kind of solid it is, and why it conducts.



Past papers

Year	Question
2023	Q23
2024	Q24



Past Paper Finder

Look back — by the end of this part...



- 1 I can describe how energy bands form from the energy levels of single atoms.
- 2 I can name and describe the conduction band and the valence band.
- 3 I can use band structure to explain why a solid conducts or does not.

2013 Higher Physics: scan the code and answer these.



Q8

Q11

Section A · charged particles,
capacitors

Show your working, not just the letter.



Scan for the paper
Marking scheme

In a semiconductor, the energy gap between the valence band and the conduction band is



- A** small, allowing some conduction to take place at room temperature
- B** large, allowing some conduction to take place at room temperature
- C** zero, allowing electrons to move freely
- D** large, meaning that no conduction can take place at room temperature
- E** small, meaning that no conduction can take place at room temperature.

Show Me: classify the solid. Answer on your board, one at a time.



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State which band an electron must reach before it can carry a current.

2

A solid has a full valence band and a very large gap. State what kind of solid it is, and why it does not conduct.

3

State the two things a solid needs before it can conduct.

Show Me: classify the solid. Answer on your board, one at a time.



- 4 Explain why a semiconductor conducts better when it is warmed up.
- 5 A solid's conduction band and valence band overlap. State what kind of solid it is, and why it conducts.

PART 2

Doping

How can adding a tiny amount of impurity make a semiconductor conduct better?

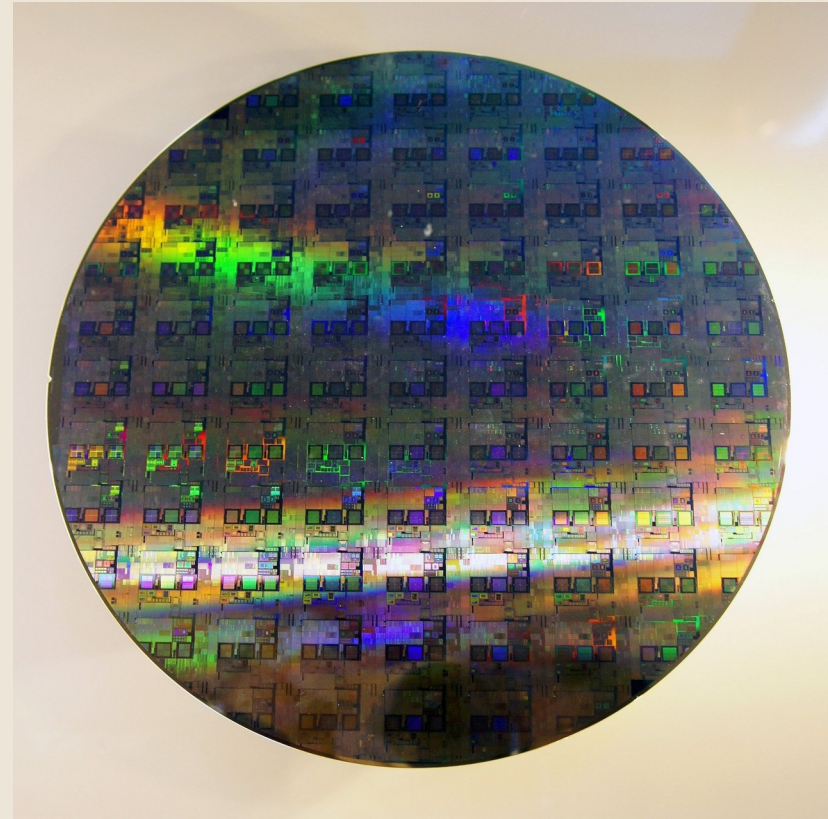
By the end of this part...



- 1 I can describe what doping is and state why it is done.
- 2 I can tell p-type and n-type semiconductors apart.
- 3 I can explain both types using their band structure.

What doping is

- Pure silicon conducts very little at room temperature.
- Chip makers add a tiny amount of impurity.
- Chips start as doped wafers like this one.

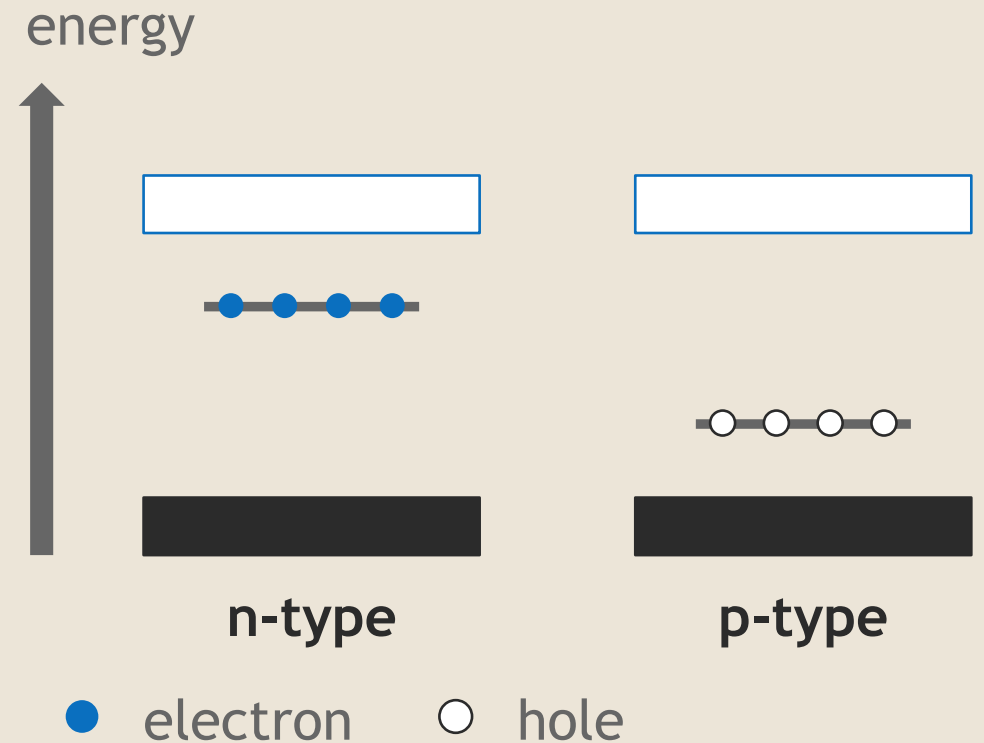


Doping



Doping means adding a small amount of a specific impurity to a semiconductor during manufacture.

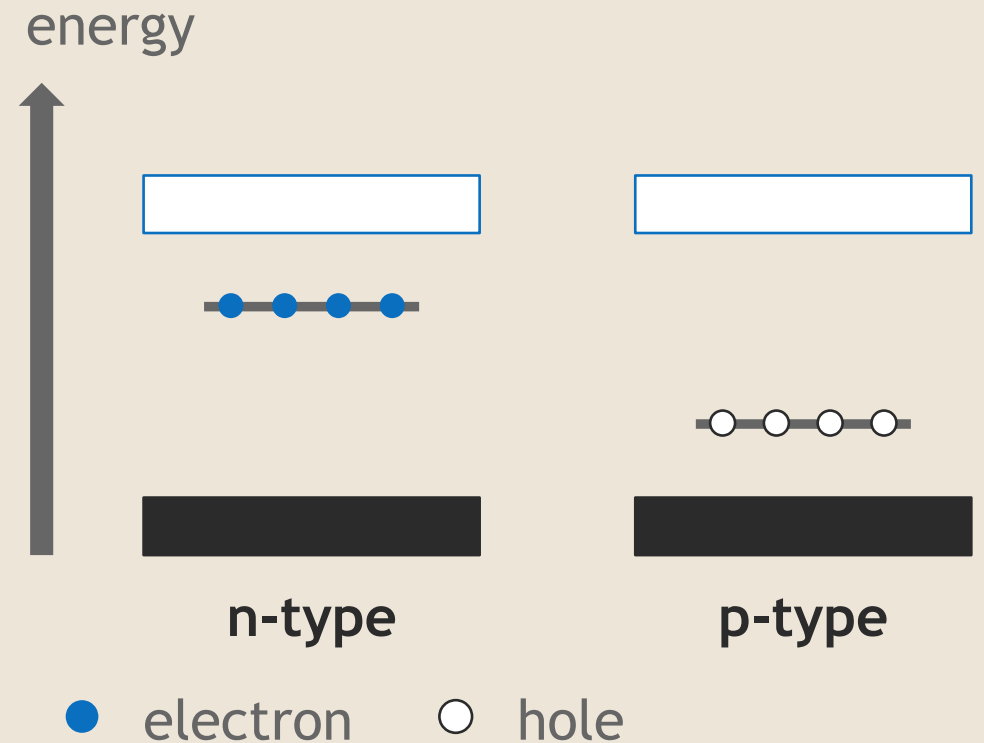
It is done to increase the conductivity in a controlled way.



Two types of doped semiconductor



Doping produces two types of semiconductor: n-type and p-type.

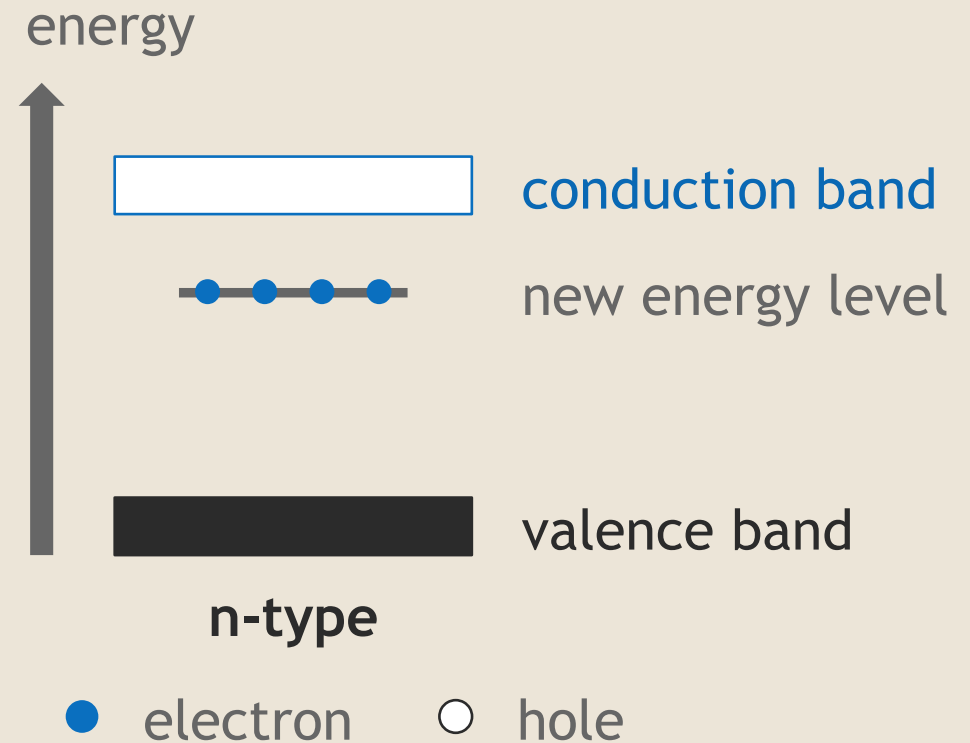


n-type semiconductors



In an n-type semiconductor the new level sits just below the conduction band.

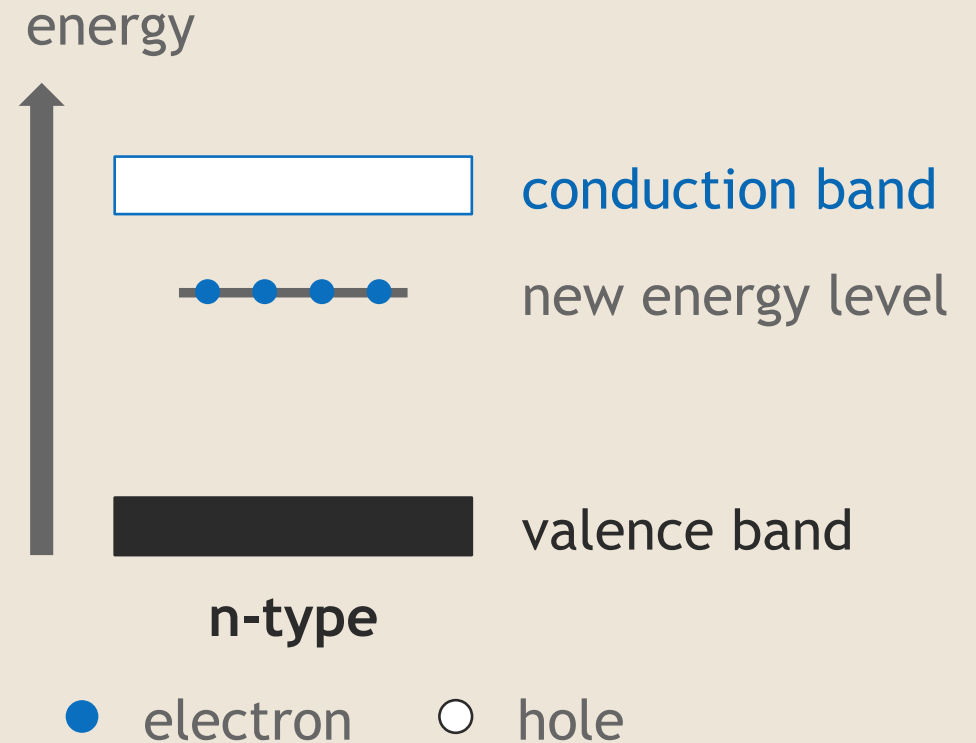
Electrons on that level need only a little energy to reach the conduction band.



Conduction in n-type



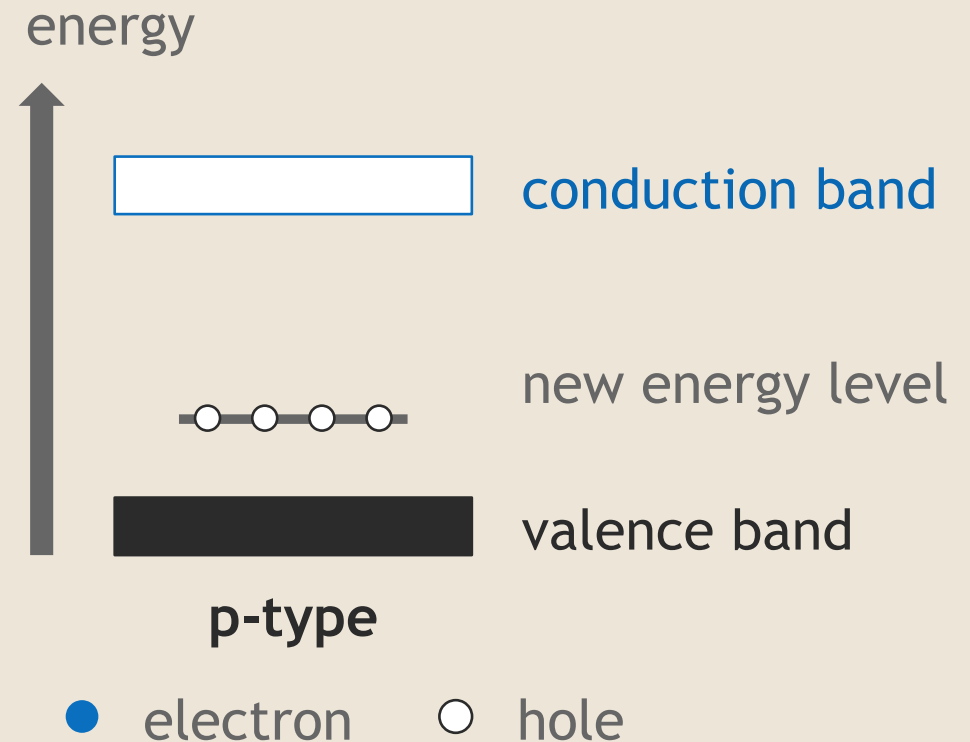
So conduction happens by electrons in the conduction band, and the charge carriers are negative.



p-type semiconductors



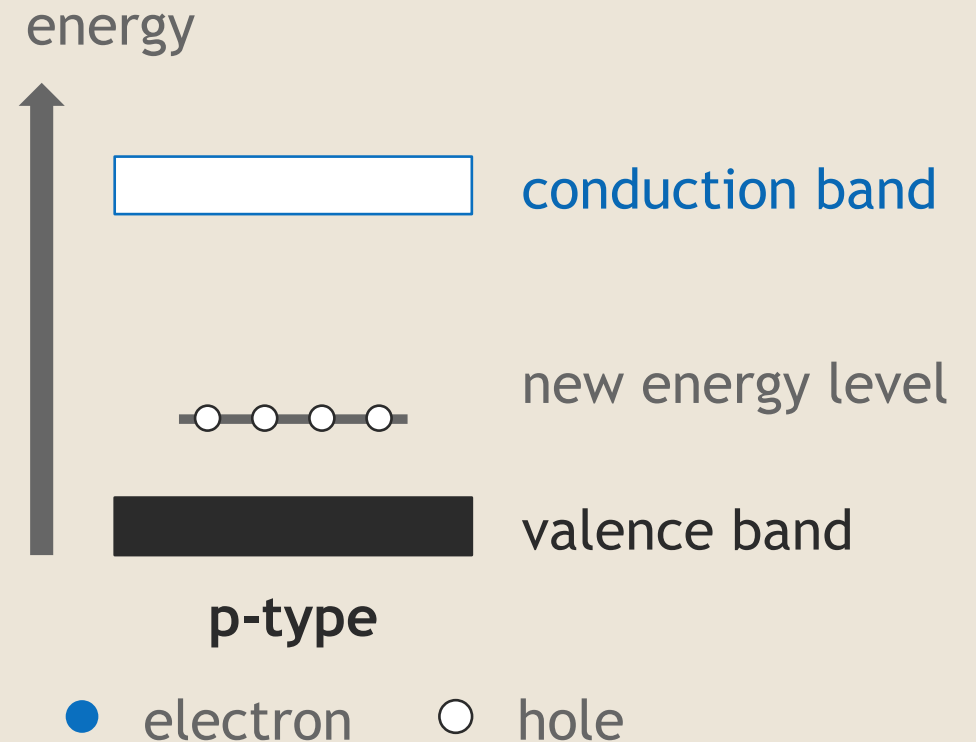
In a p-type semiconductor the new level sits just above the valence band, and it is empty. Electrons in the valence band move up into those empty states, leaving holes behind them.



Conduction in p-type



So conduction happens in the valence band, and the charge carriers are holes, which behave as if they are positive.



A crystal of silicon is “doped” with arsenic. This means that a small number of the silicon atoms are replaced with arsenic atoms.



The effect of the doping on the crystal is to

- A make it into a photodiode
- B make it into an insulator
- C increase its resistance
- D decrease its resistance**
- E allow it to conduct in only one direction.

Show Me: doping. Answer on your board, one at a time.



- 1 State what is meant by doping a semiconductor.
- 2 State why a semiconductor is doped.
- 3 A semiconductor's charge carriers are electrons. State whether it is p-type or n-type.

Show Me: doping. Answer on your board, one at a time.



4

State where the new energy level sits in an n-type semiconductor.

5

Explain, in terms of bands, how conduction happens in a p-type semiconductor.



Past papers

Year	Question
2011	Q18
2014	Q18



Past Paper Finder

Look back — by the end of this part...



- 1 I can describe what doping is and state why it is done.
- 2 I can tell p-type and n-type semiconductors apart.
- 3 I can explain both types using their band structure.

2015 Revised Higher Physics: scan the code and answer these.



Q17

Q19

Q20

Section A · a.c., semiconductors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

PART 3

The p-n junction

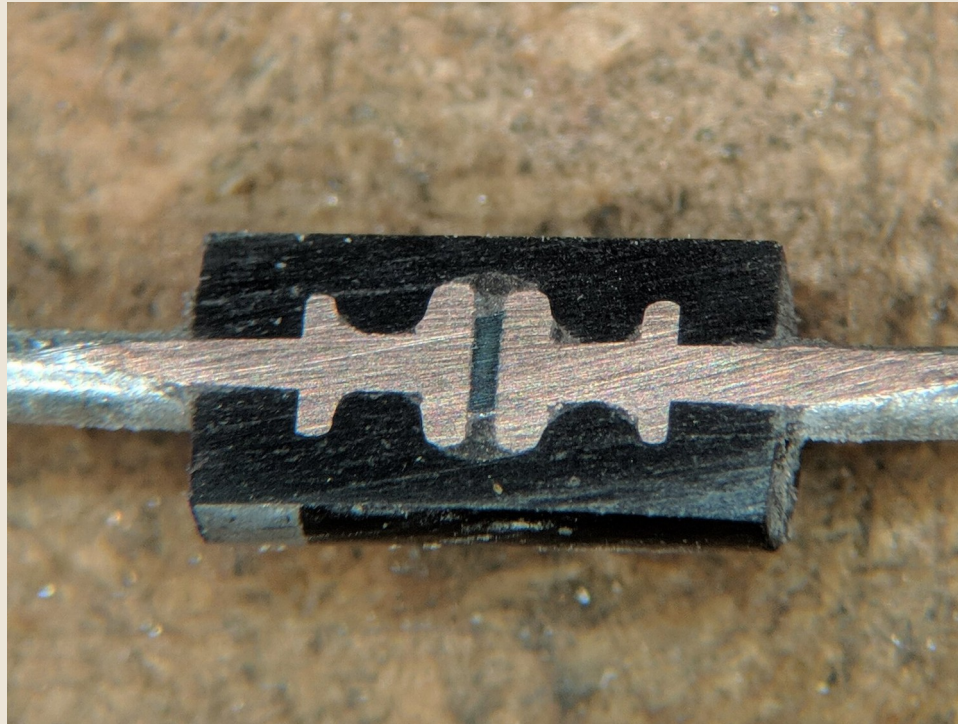
What happens where p-type meets n-type, and why does a diode pass current only one way?

By the end of this part...



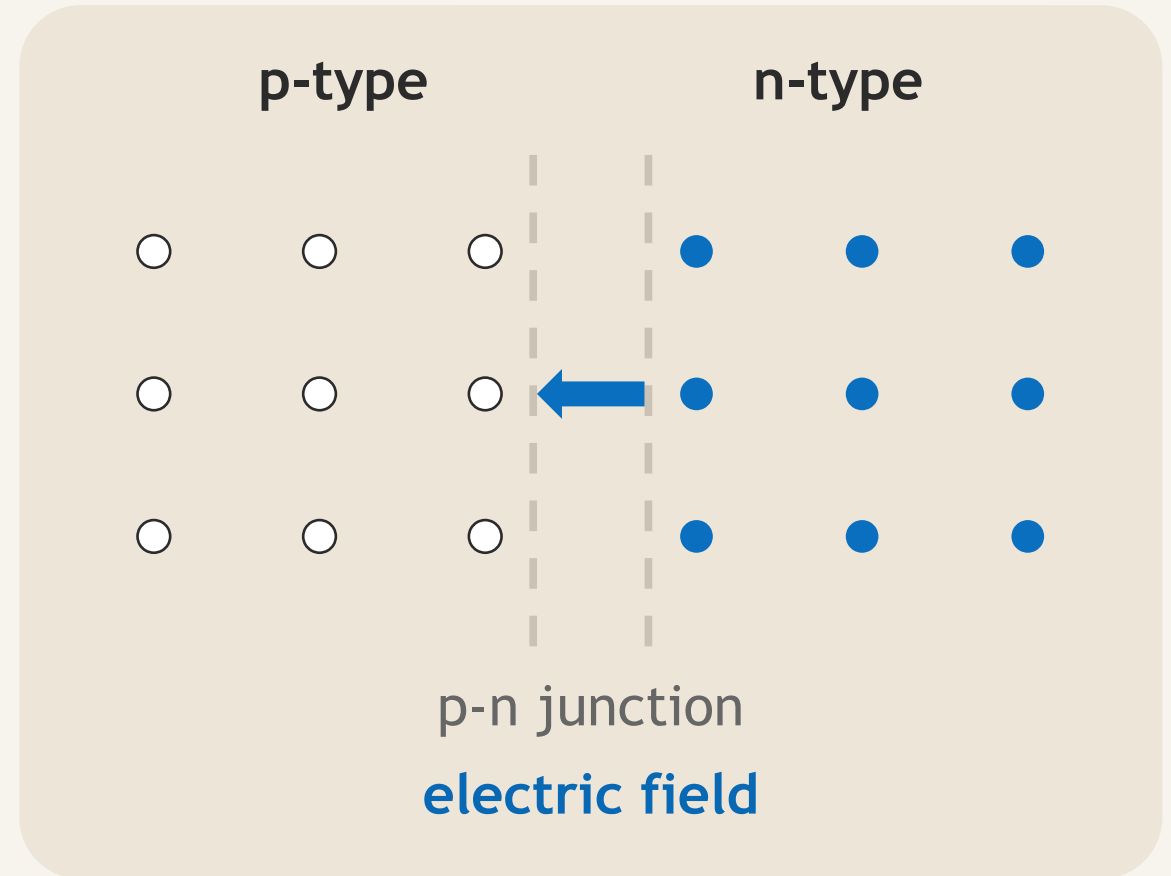
- 1 I can describe how a p-n junction is formed and what it contains.
- 2 I can state what forward bias and reverse bias mean.
- 3 I can state the effect of each on the electric field in the junction.

A diode cut in half. Where is its p-n junction?



How the junction forms

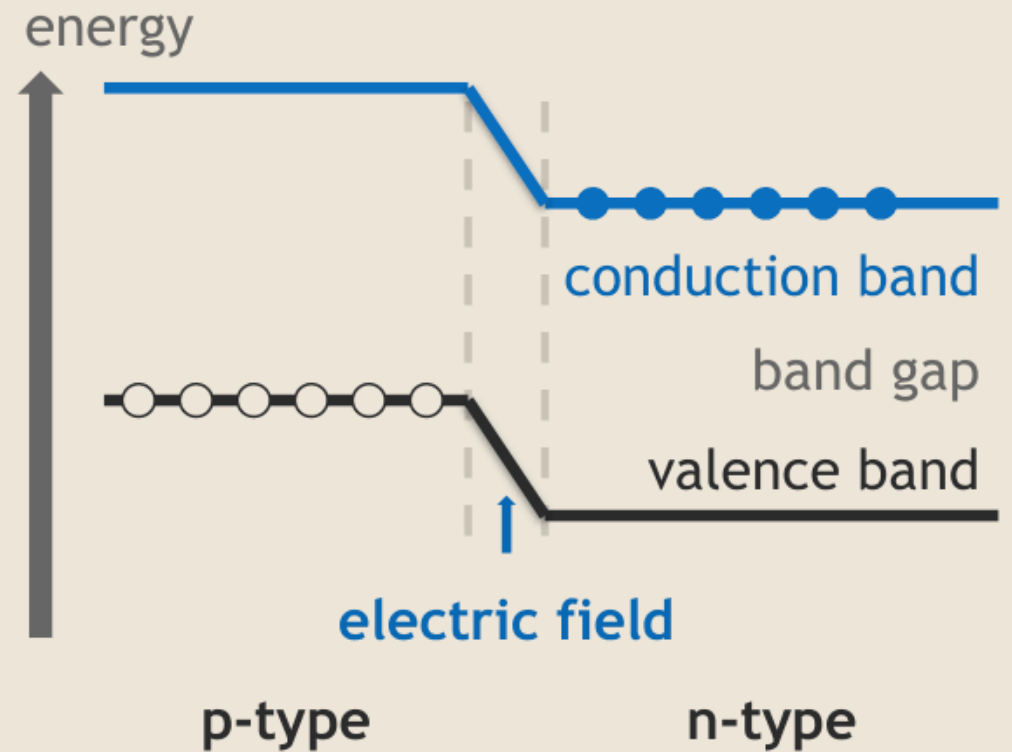
- One piece: p-type at one end, n-type at the other.
- Where the two layers meet is the p-n junction.
- The junction contains an electric field.



The p-n junction



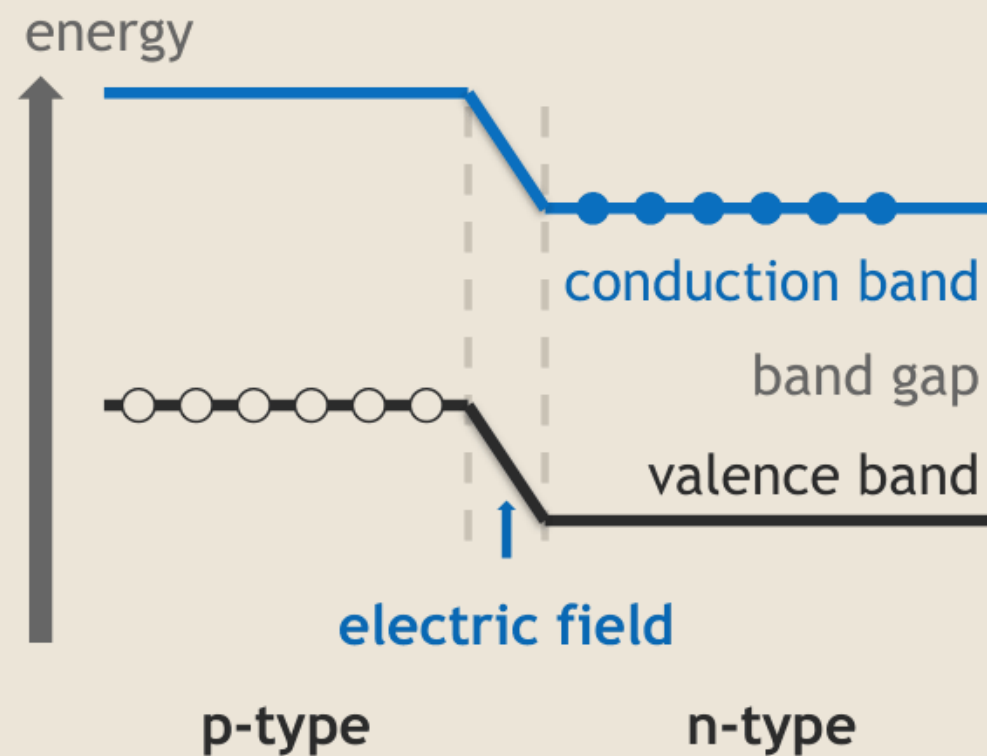
A p-n junction is formed where a p-type layer and an n-type layer sit next to each other. There is an electric field in the p-n junction.



The field shows up as a step



On a band diagram that field shows up as a step: both bands drop as you cross from the p-type side to the n-type side.

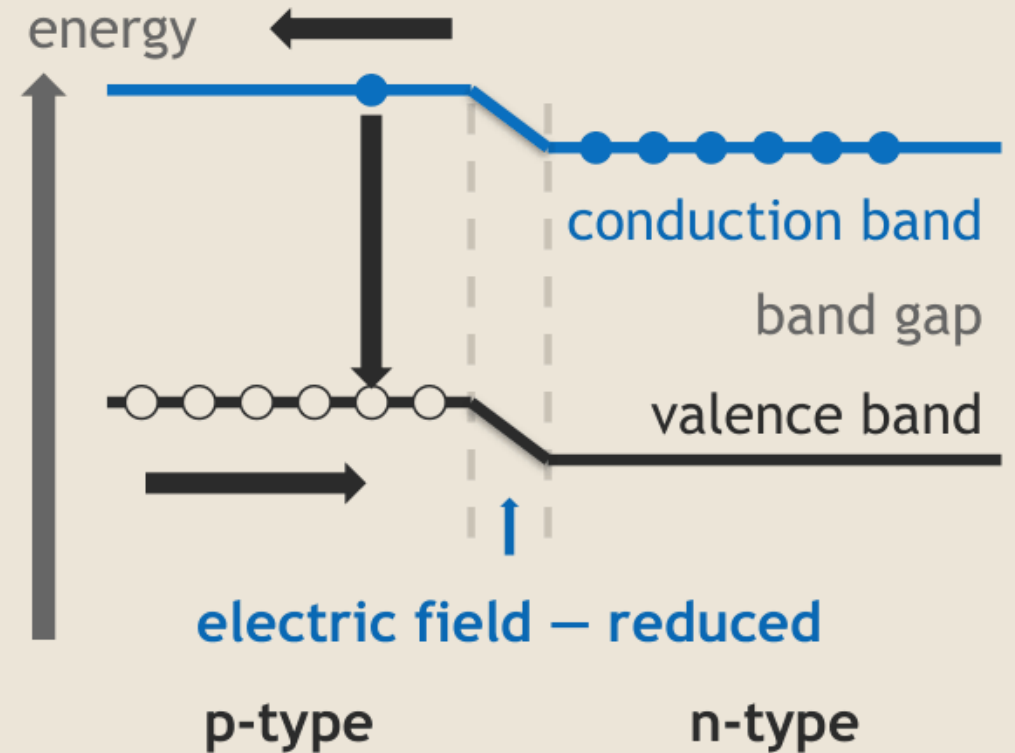


Forward bias



Forward bias reduces the electric field in the junction, so the step gets smaller.

An electron starts in the conduction band of the n-type, moves to the conduction band of the p-type, then falls to the valence band of the p-type. A current flows.

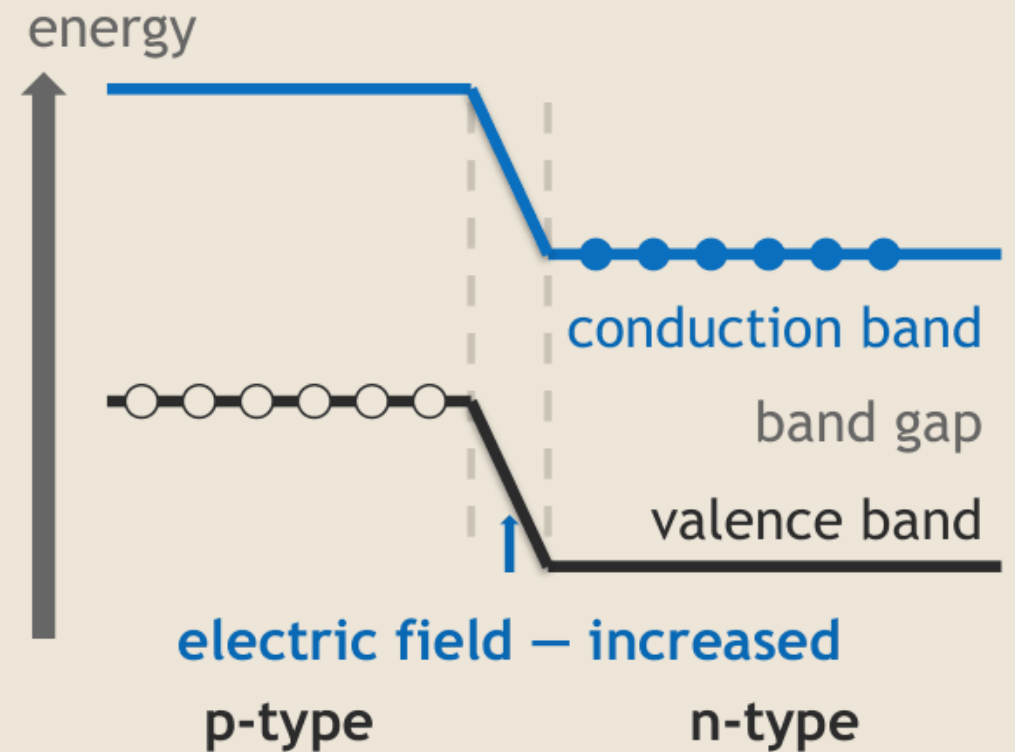


Reverse bias



Reverse bias increases the electric field in the junction, so the step gets bigger.

An electron in the conduction band of the n-type cannot move to the conduction band of the p-type. No current flows.



Spotting the bias in a circuit



The triangle in the diode symbol points the way the current is allowed to flow.

If the triangle points from the positive side towards the negative side, the diode is forward biased and it conducts.



forward biased: conducts



reverse biased: no current

When a diode is reverse biased



Pointing the other way, it is reverse biased and nothing flows.

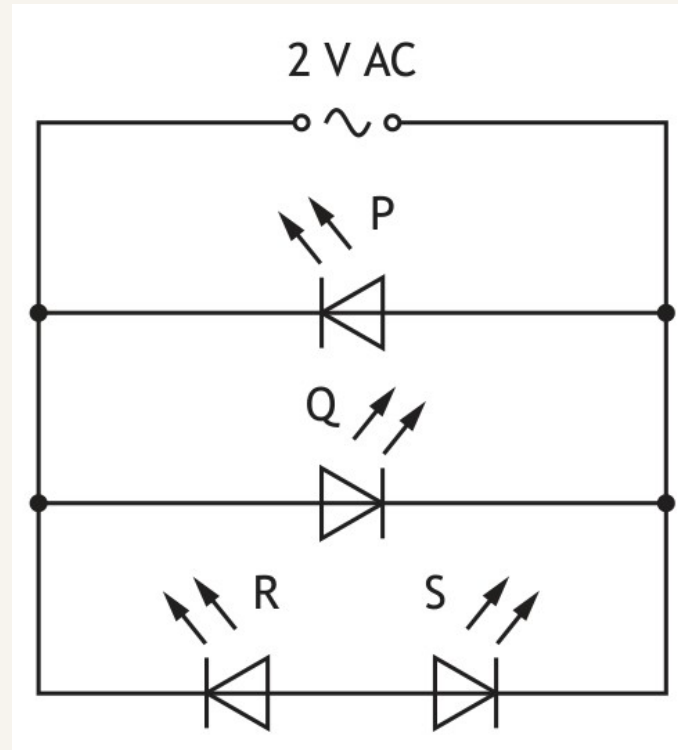


forward biased: conducts

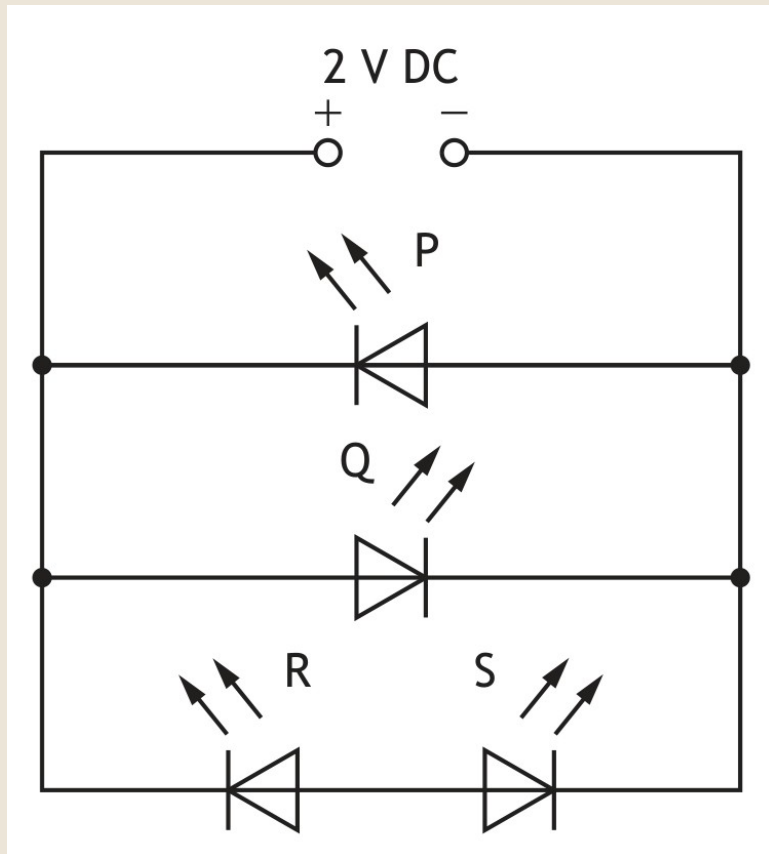


reverse biased: no current

Four LEDs on one supply. Which ones are forward biased?



A student connects four identical light emitting diodes (LEDs) to a 2 V DC supply as shown.



Which of the LEDs P, Q, R, and S will light?

A P only

B Q only

C P and Q only

D P and R only

E Q and S only

Show Me: junctions and bias. Answer on your board, one at a time.



1

State what is produced when p-type and n-type layers sit next to each other.

2

State what a p-n junction contains.

3

State the effect of forward bias on the electric field in a junction.

Show Me: junctions and bias. Answer on your board, one at a time.



4 State the effect of reverse bias on the electric field in a junction.

5 A diode is connected so that no current flows. State how it is biased, and what has happened to the field.



Past papers

Year	Question
2023	Q5(b)(ii)



Past Paper Finder

Look back — by the end of this part...



- 1 I can describe how a p-n junction is formed and what it contains.
- 2 I can state what forward bias and reverse bias mean.
- 3 I can state the effect of each on the electric field in the junction.

2014 Higher Physics: scan the code and answer these.



Q10

Q11

Section A · a.c., capacitors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

PART 4

Light-emitting diodes

An LED is a forward-biased p-n junction doing one very specific job: turning electrons into photons.

By the end of this part...



- 1 I can describe an LED as a forward-biased p-n junction diode.
- 2 I can describe, in terms of bands, how an LED emits a photon.
- 3 I can explain why different LEDs emit different colours.

What an LED is

- An LED is a p-n junction diode that gives out light.
- It lights only when it is forward biased.
- Traffic lights, screens and street lamps all use them.

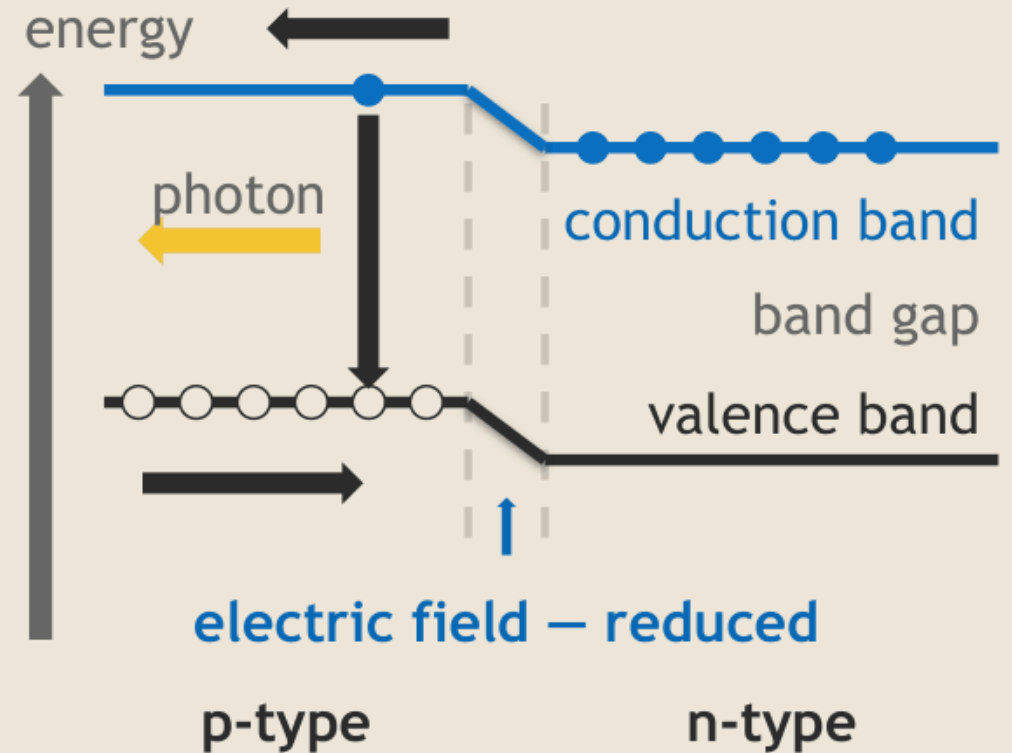


Light-emitting diodes



An LED is a forward-biased p-n junction diode that emits photons.

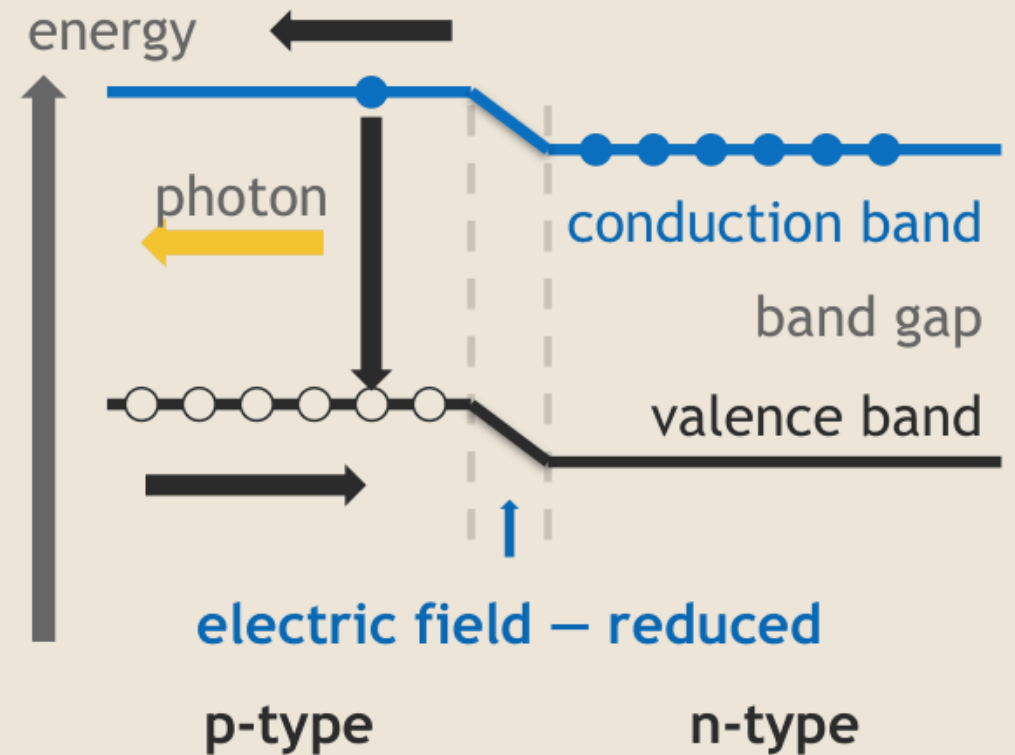
An electron starts in the conduction band of the n-type and moves to the conduction band of the p-type.



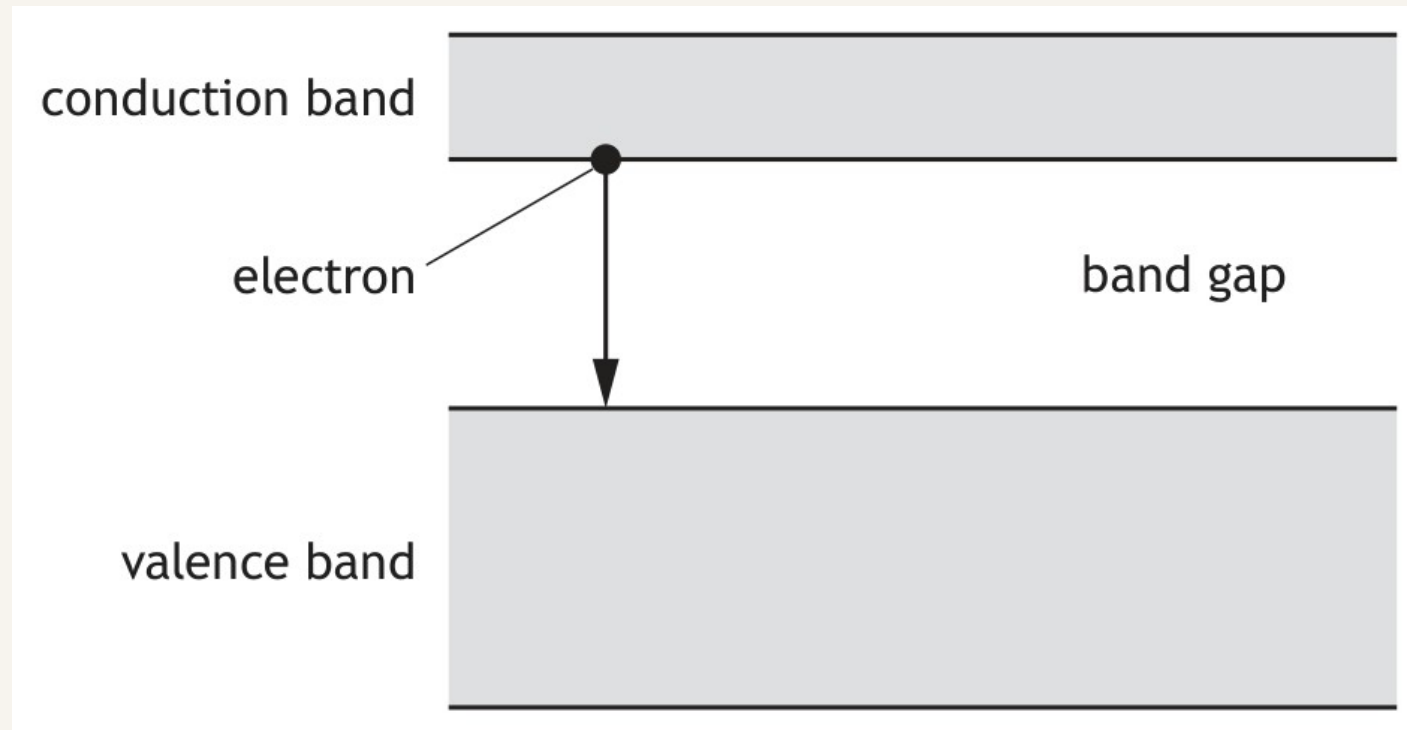
How an LED emits a photon



The electron then falls from the conduction band to the valence band of the p-type. As it falls, it emits a photon.



In an exam: one electron, one drop, one photon

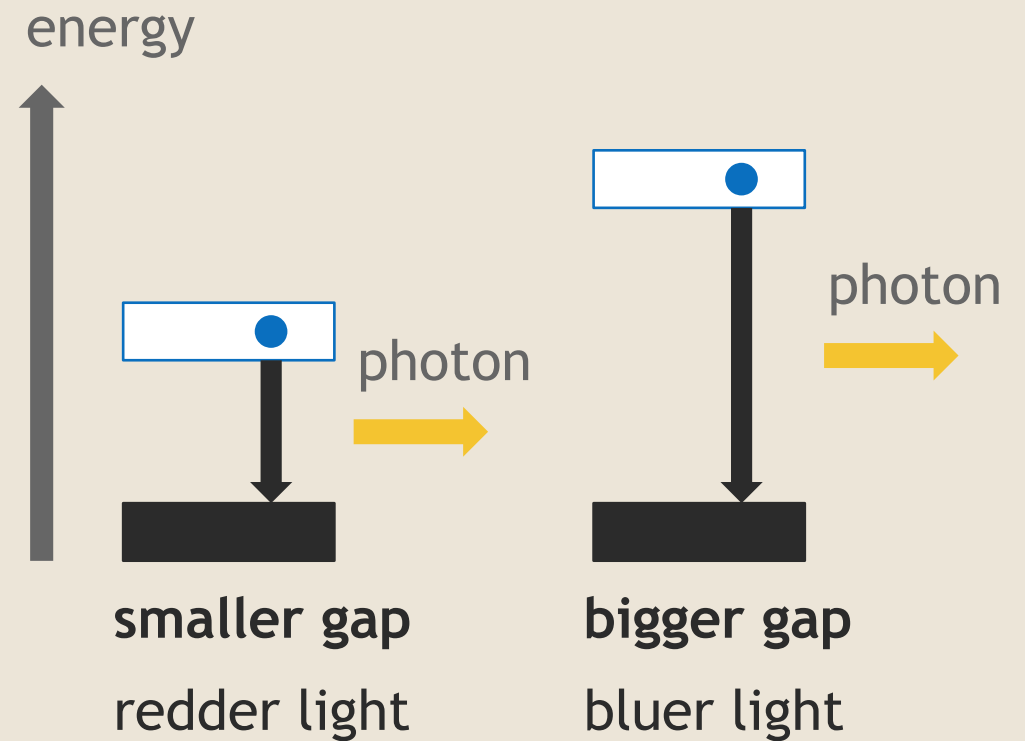


Band gap and colour



The electron falls from the conduction band to the valence band of the p-type, so it loses exactly the energy of the band gap.

All of that energy leaves as one photon, so the photon's energy is equal to the band gap.

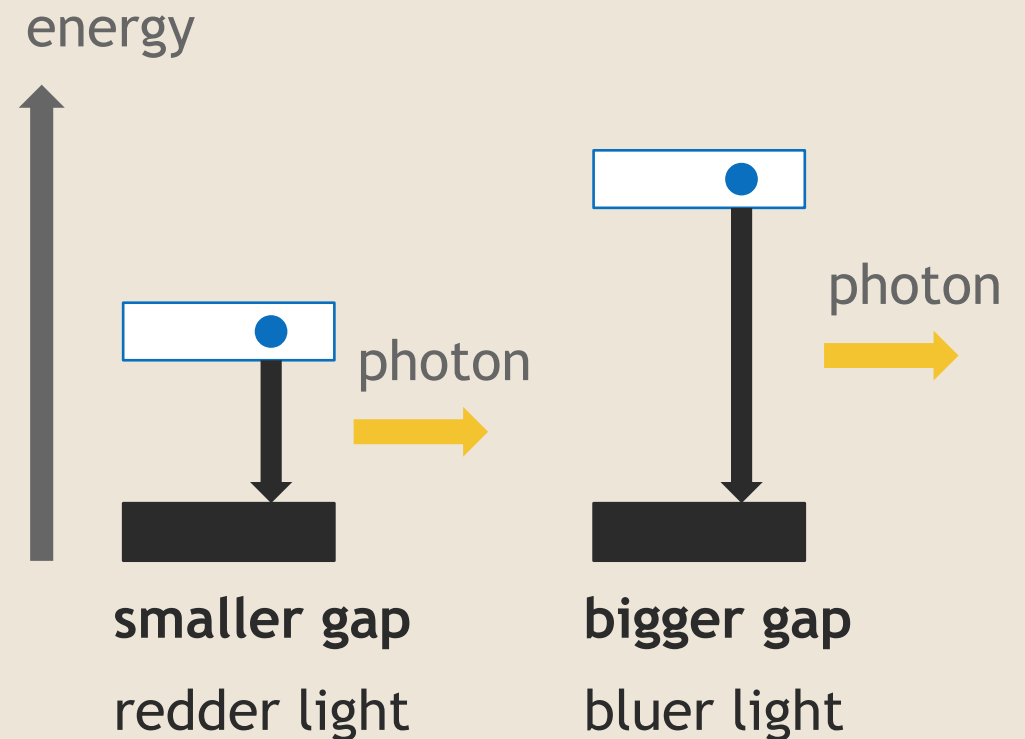


Bigger gap, bluer light



An electron that falls across a bigger band gap emits a higher-energy photon: bluer light.

An electron that falls across a smaller band gap emits a lower-energy photon: redder light.



A student makes the following statements about red and blue LEDs.



- I Red LEDs emit photons of longer wavelength than blue LEDs.
- II The gap between the valence band and the conduction band is larger for red LEDs than blue LEDs.
- III The potential difference across an LED determines the wavelength of the photons emitted.

Which of these statements is/are correct?

A I only

B II only

C III only

D I and II only

E I, II and III

Watch Out — always mention the band gap

THE MISTAKE

- Explaining LED colour without mentioning the band gap
- Treating every LED's photon energy as the same

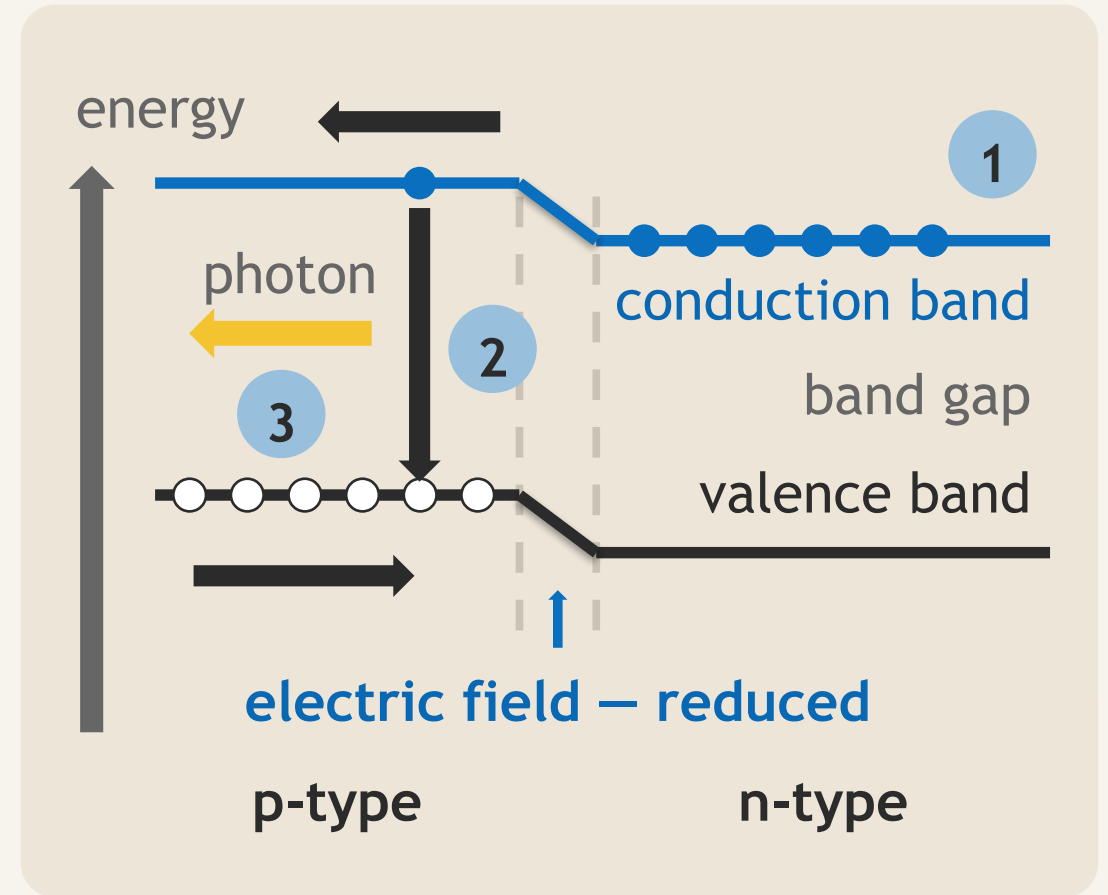
THE CORRECT ANSWER

- Photon energy equals the band gap. Say so.
- A different colour means a different gap.

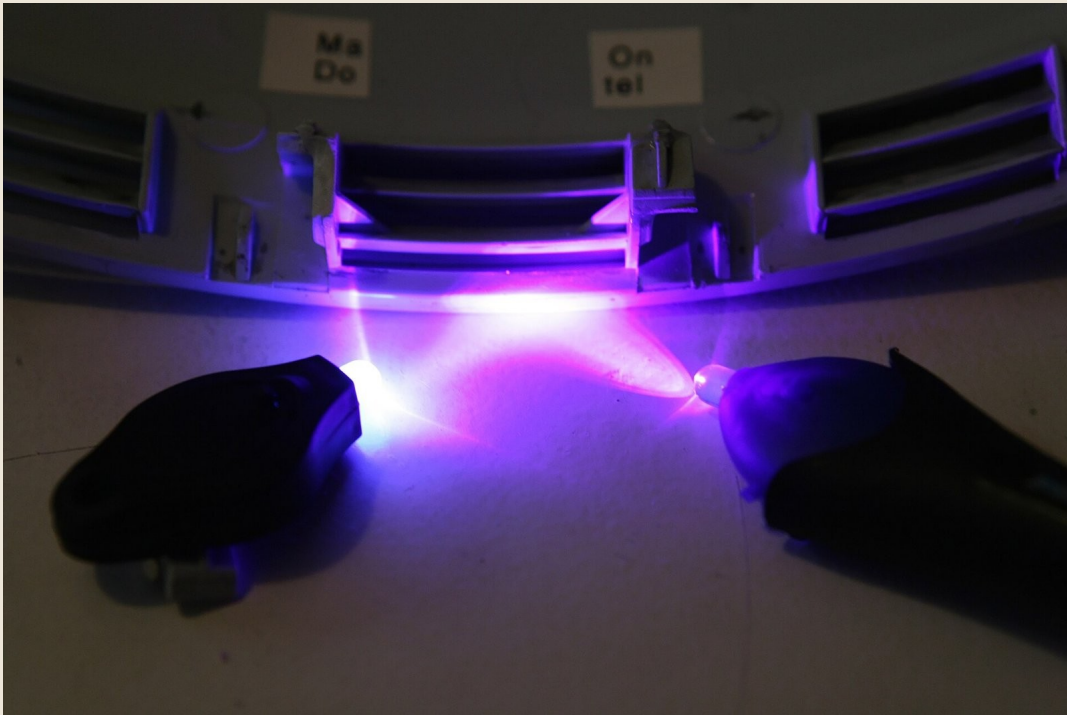
A 3-mark LED answer: what each mark needs

- 1 Electrons move from the n-type conduction band to the p-type conduction band.
- 2 They fall to the p-type valence band.
- 3 A photon is emitted.

Name the bands in full. “Holes and electrons recombine” alone scores 0.



Which of these LEDs has the bigger band gap? How do you know?



ultraviolet LEDs curing glue



an LED street lamp

Show Me: LEDs. Answer on your board, one at a time.



- 1 State what an LED is, in terms of a p-n junction.
- 2 State how an LED must be biased before it will emit light.
- 3 Describe, in terms of bands, how an LED emits a photon.

Show Me: LEDs. Answer on your board, one at a time.



4

Two LEDs emit red and blue light. State and justify which has the larger band gap.

5

Explain why one LED emits a single colour rather than a range of colours.



Past papers

Year	Question
2025	Q23
2026	Q7(c)



Past Paper Finder

Look back — by the end of this part...



- 1 I can describe an LED as a forward-biased p-n junction diode.
- 2 I can describe, in terms of bands, how an LED emits a photon.
- 3 I can explain why different LEDs emit different colours.

2014 Revised Higher Physics: scan the code and answer these.



Q8

Q16

Q18

Section A · particles, semiconductors
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

PART 5

Solar cells

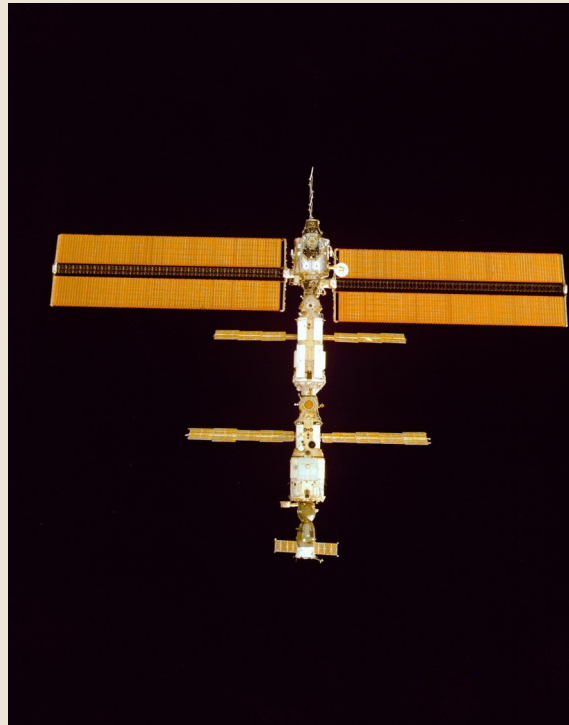
Run the LED backwards — light goes in instead of coming out — and you have a solar cell.

By the end of this part...



- 1 I can describe a solar cell as a p-n junction and name the photovoltaic effect.
- 2 I can describe, in terms of bands, how a solar cell produces a p.d.
- 3 I can state the difference between a solar cell and an LED.

The Space Station runs on solar cells. What happens inside one?



The photovoltaic effect

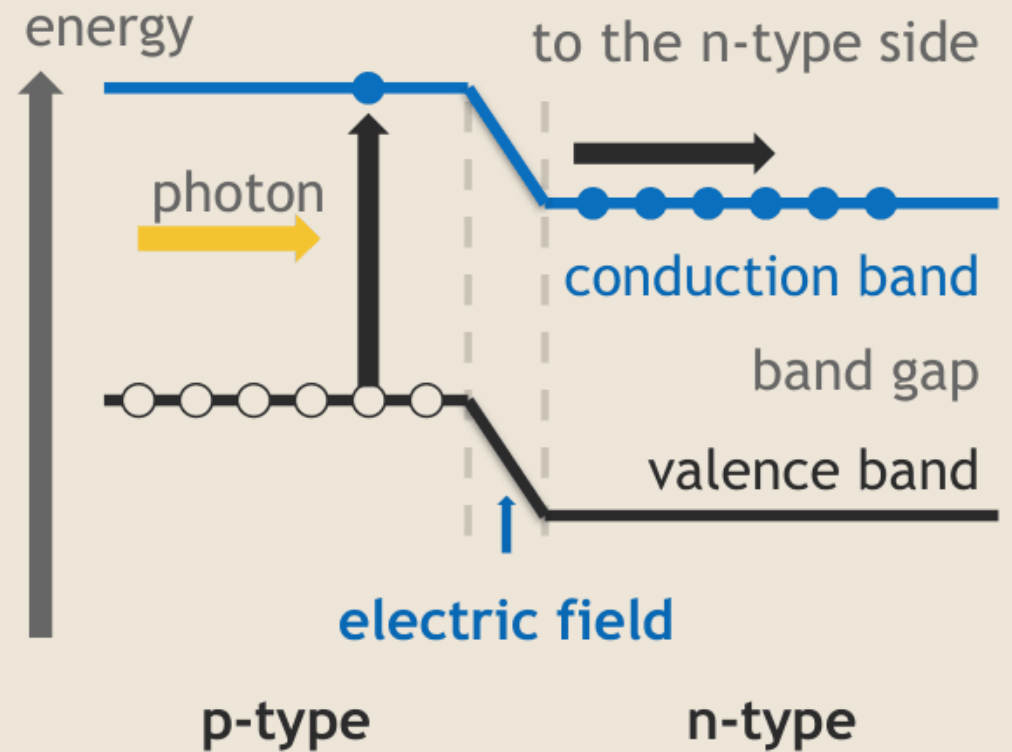
- A solar cell turns light into electrical energy.
- It is a p-n junction, just like an LED.
- Panels join many cells, like these in Midlothian.



Solar cells



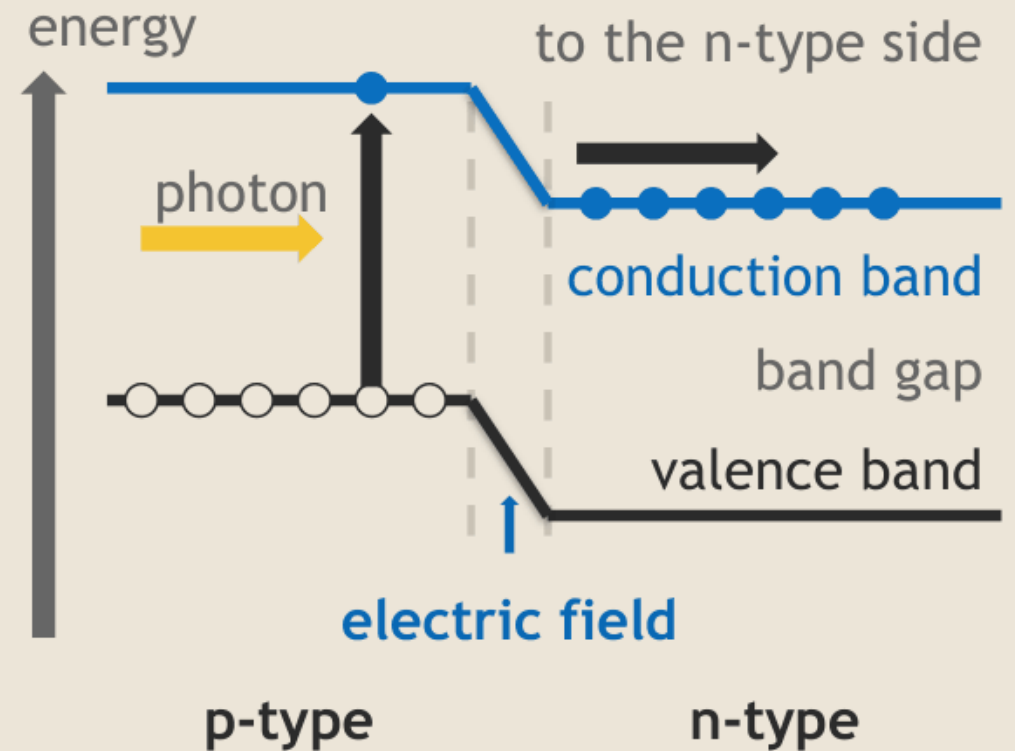
A solar cell is a p-n junction designed to produce a potential difference when photons are absorbed. This is called the photovoltaic effect.



From photon to moving electron



An electron in the valence band of the p-type absorbs a photon and rises to the conduction band of the p-type. The junction's electric field then moves it to the conduction band of the n-type.

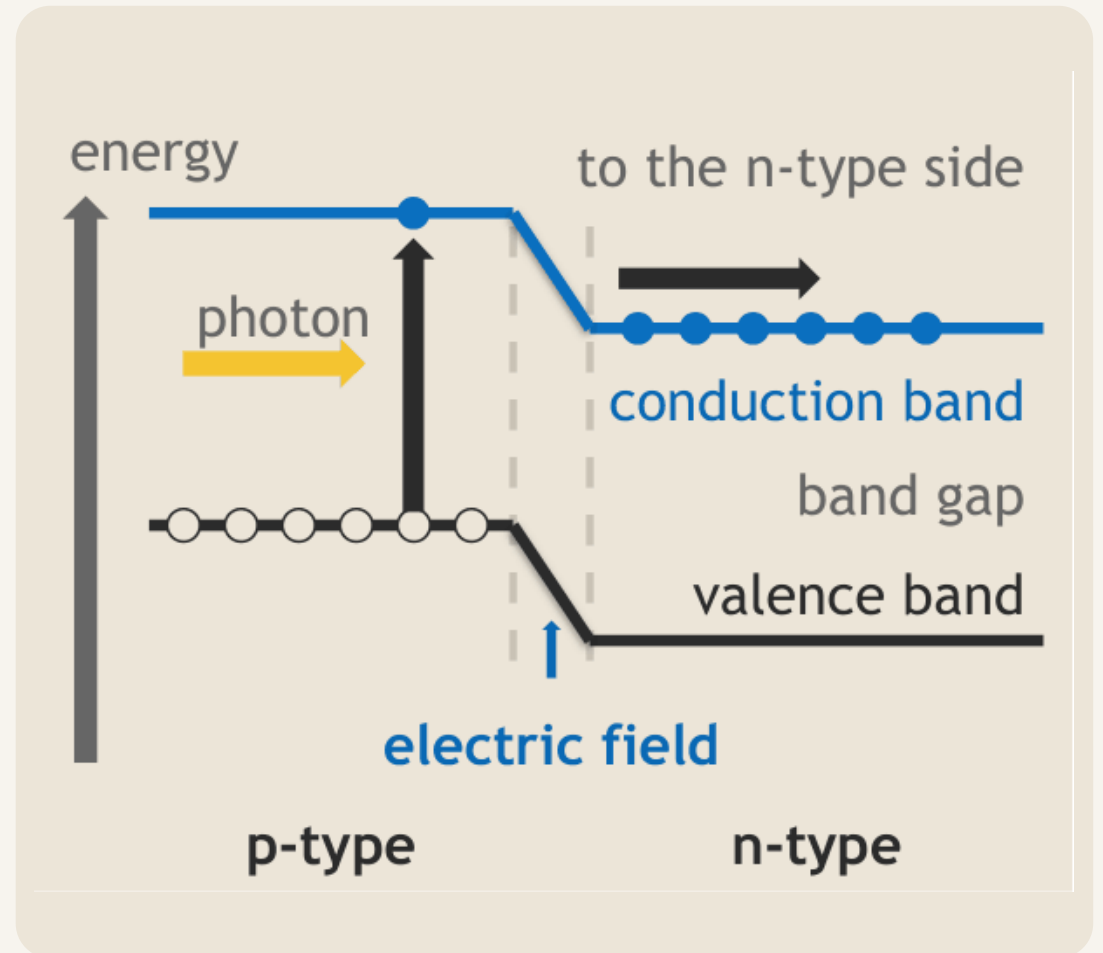


How the potential difference appears



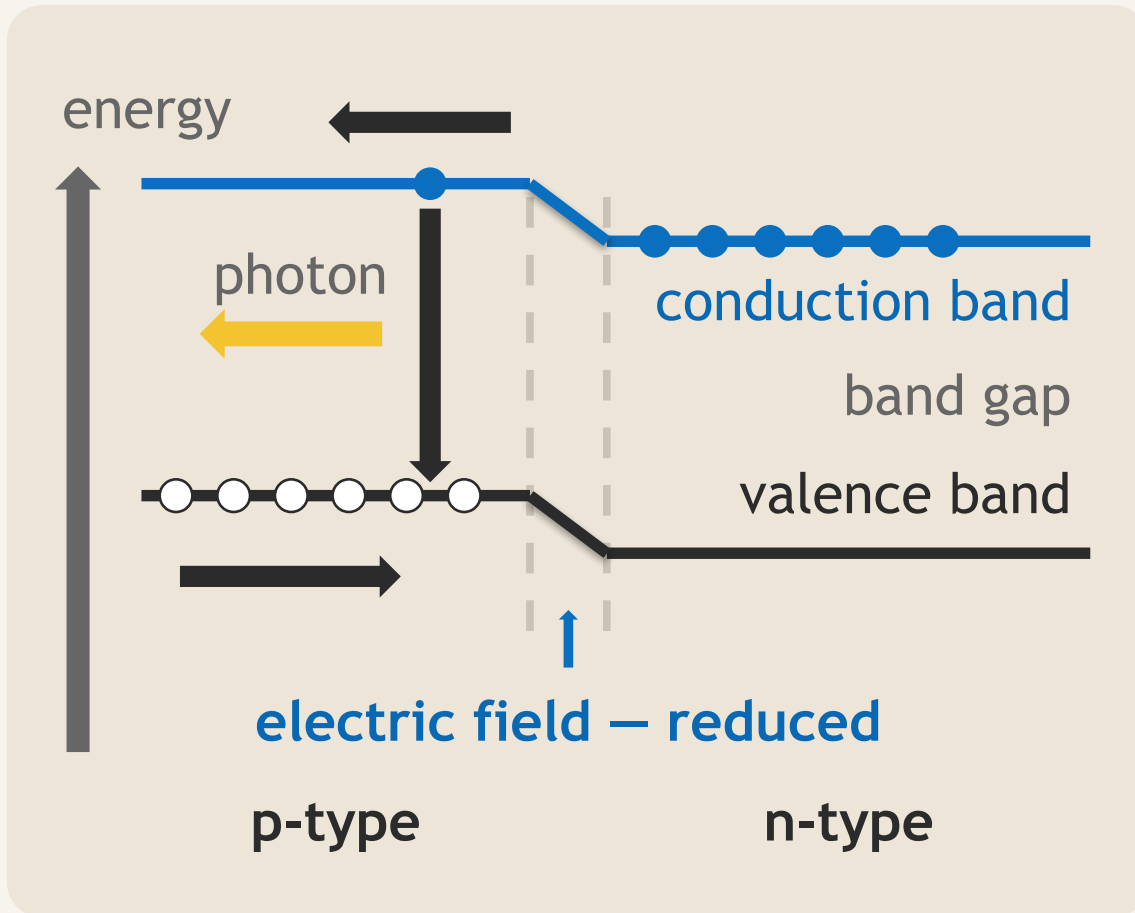
Electrons gather in the conduction band of the n-type, and holes in the valence band of the p-type. This produces a potential difference.

That potential difference appears across the whole solar cell.

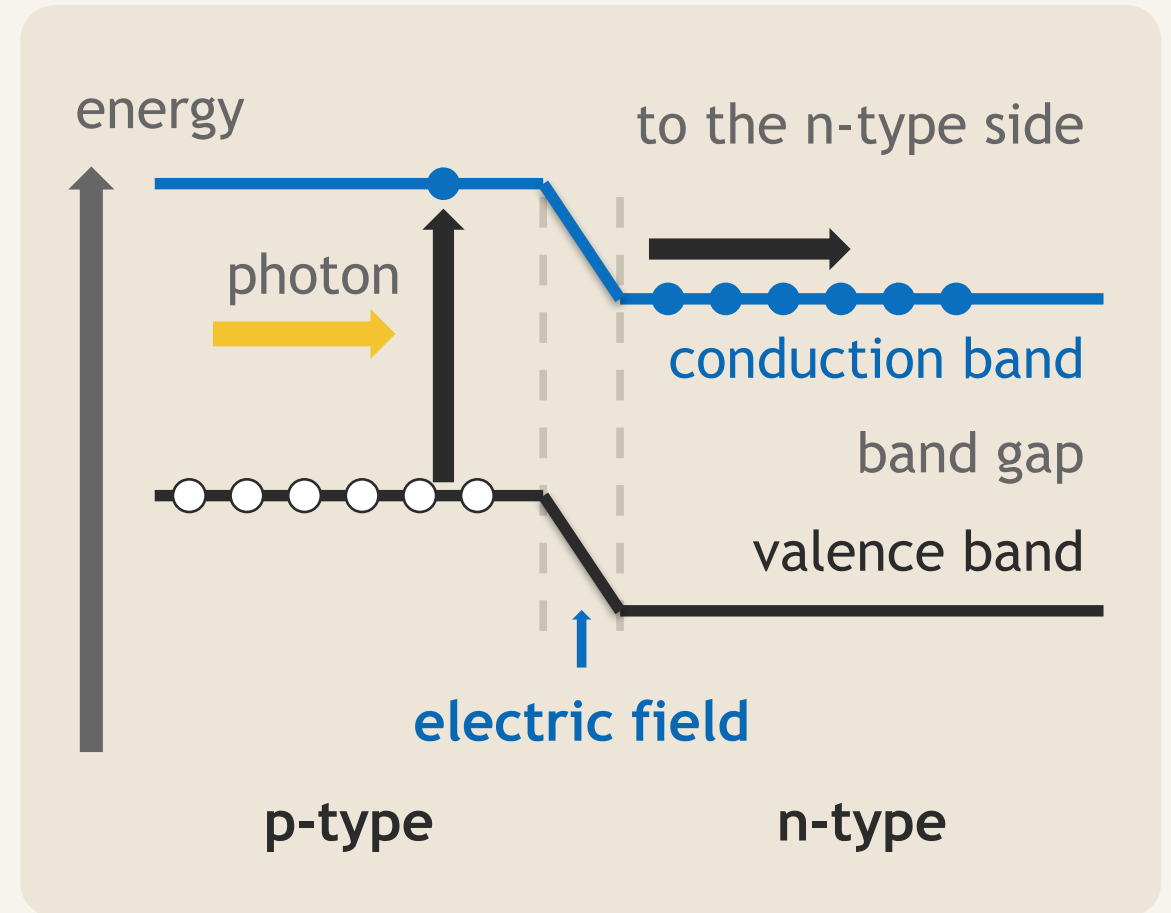


LED or solar cell?

LED: photon out



Solar cell: photon in



Watch Out — solar cell, not LED

THE MISTAKE

- Describing a solar cell as if it emits photons
- Mixing up which device absorbs and which emits

THE CORRECT ANSWER

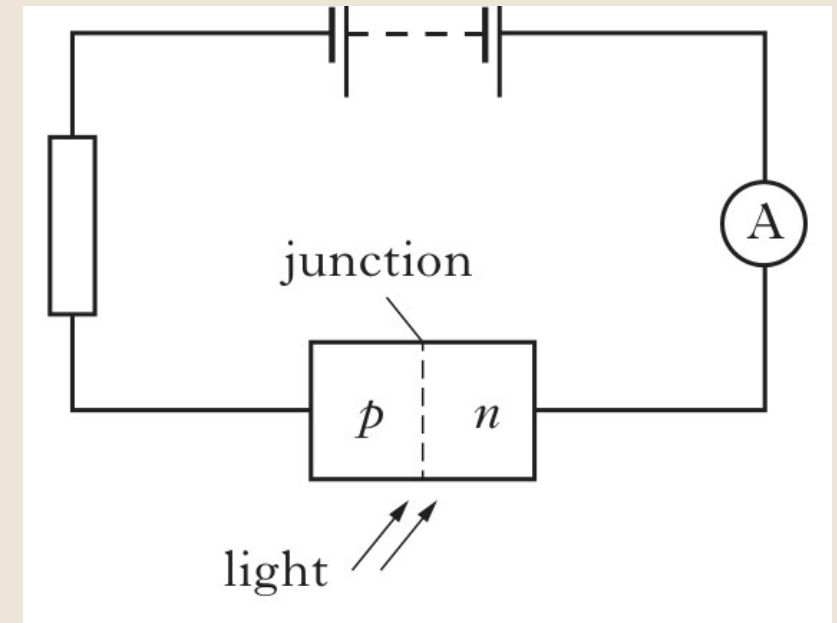
- A solar cell ABSORBS a photon; an LED EMITS one.
- Check which device the question actually names.

Reading the circuit version



The box marked p and n is the junction itself, with the dashed line showing where the two layers meet. The arrows show light falling on the junction.

Light in gives a potential difference out, which is what the meter in the circuit reads.



A student makes the following statements about p-n junction devices.



- I In solar cells, a potential difference is produced when photons are incident on the junction.
- II The photovoltaic effect occurs in solar cells.
- III In LEDs, photons are emitted from the junction when a current is passed through it.

Which of these statements is/are correct?

A I only

B III only

C I and II only

D I and III only

E I, II and III

Show Me: solar cells. Answer on your board, one at a time.



- 1 Name the effect by which a solar cell produces a potential difference.
- 2 State what happens to an electron when a solar cell absorbs a photon.
- 3 Describe, in terms of bands and the junction's field, how a solar cell produces a p.d.

Show Me: solar cells. Answer on your board, one at a time.



- 4 A solar cell and an LED are both p-n junctions. State the key difference in what each one does.
- 5 Explain why a solar cell produces no potential difference in the dark.



Past papers

Year	Question
2017	Q14(b)
2024	Q15(b)



Past Paper Finder

Look back — by the end of this part...



- 1 I can describe a solar cell as a p-n junction and name the photovoltaic effect.
- 2 I can describe, in terms of bands, how a solar cell produces a p.d.
- 3 I can state the difference between a solar cell and an LED.

Experiments you could try

- Put an LED and a resistor across a cell, both ways round: it lights only when forward biased.
- Shine a torch on a red LED connected to a voltmeter: it gives a small p.d.
- Connect a solar cell to a voltmeter under a lamp, then cover it.
- Warm a thermistor in your hand and watch its resistance fall.

Checkpoint

- 1 State the difference between the valence band and the conduction band.
- 2 A solid's highest occupied band is only partly full. State what kind of solid it is, and why it conducts.
- 3 Explain, in terms of band structure, why a semiconductor conducts better when it is warmer.

Checkpoint

- 4 State what doping is, and where it puts a new energy level.
- 5 Describe, in terms of bands, how an LED emits a photon.



Past papers, Paper 2

Year	Question	What it asks
2024	Q15(b)	solar cell, not an LED
2018	Q11(c)	red LED vs blue LED
2026	Q7(c)	doping and the LED

LOOKING AHEAD

Next: Motion, Graphs & Projectiles

Scalars, vectors, and the equations that connect them – the largest topic of the course.

Plenary. Answer on your board, one at a time.



- 1 Name the two bands, and say what the gap between them means.
- 2 Explain, using band structure, why an insulator does not conduct.
- 3 State what doping is, and why a semiconductor is doped.

Plenary. Answer on your board, one at a time.



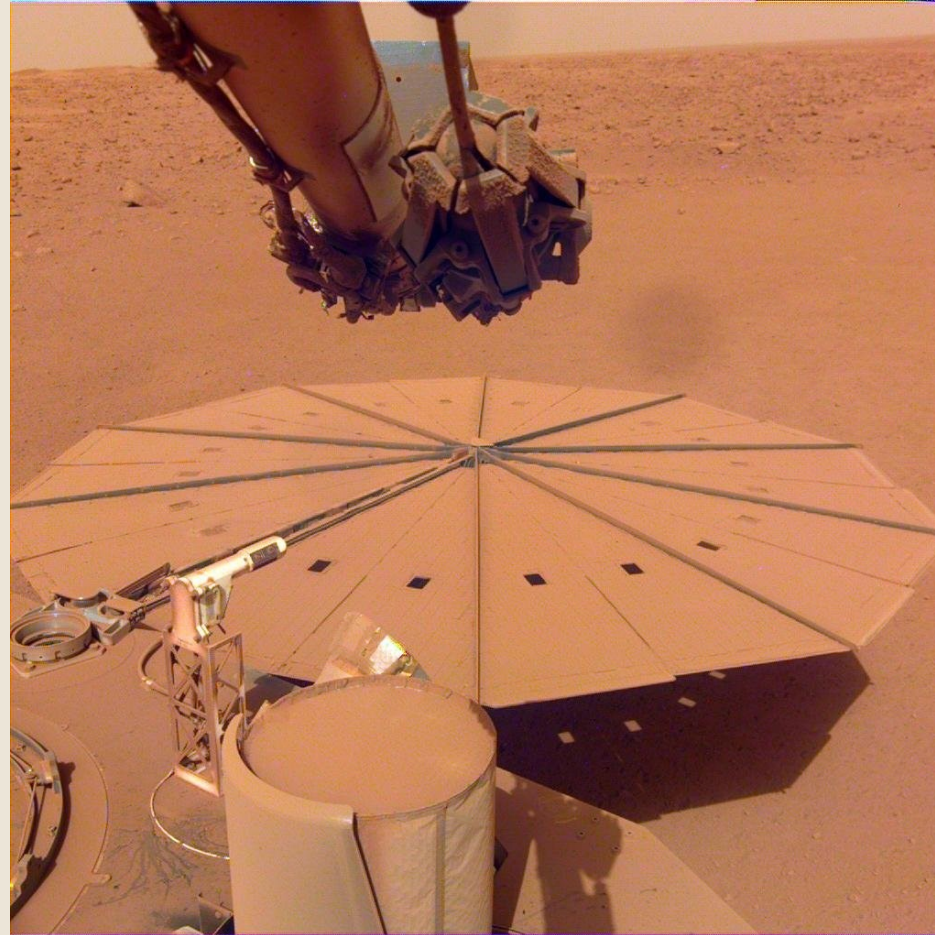
- 4 State the effect of forward bias on the field in a p-n junction.
- 5 An LED emits light and a solar cell absorbs it. Explain the difference in terms of the bands.

Extension. Answer on your board, one at a time.



- 1 A pure semiconductor is doped n-type. Sketch its band diagram before and after, and explain what changed.
- 2 Solar panels lose efficiency as they get hot, yet semiconductors conduct better when warm. Suggest why.
- 3 Explain, using band diagrams, why a p-n junction passes current one way but not the other.

Extension, question 2: InSight's dusty solar panel on Mars.



Credits (1 of 3)

Starter (2009 Higher Physics): ©
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Which of these conducts best, and which worst?: Photo: RobertKuhlmann, CC BY-SA 3.0, Wikimedia Commons; Photo: Maoileann, CC BY-SA 4.0, Wikimedia Commons; Photo: Peellden, CC BY-SA 3.0, Wikimedia Commons

The exam diagram: bands named, not shaded:
© Qualifications Scotland (SQA) · Higher Physics
2024 P1 Q23

Hinge question: the gap in a semiconductor:
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Higher specimen QP, Q14

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What doping is: Photo: Peellden, CC BY-SA 3.0, Wikimedia Commons

Hinge question: doping silicon: ©
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2013 Q20

Credits (2 of 3)

Starter (2015 Revised Higher Physics): © Qualifications Scotland (SQA)

Inside a diode: Photo: TubeTimeUS, CC BY-SA 4.0, Wikimedia Commons

Bias in a real exam question: © Qualifications Scotland (SQA) · Higher Physics 2024 P1 Q25

Hinge question: which LEDs light?: © Qualifications Scotland (SQA) · Higher specimen Paper 1 Q20

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What an LED is: Photo: Kevin Payravi, CC BY-SA 4.0, Wikimedia Commons

In an exam: one electron, one drop, one photon: © Qualifications Scotland (SQA) · Higher 2023 P2 Q5(b)(iii)

Hinge question: red and blue LEDs: © Qualifications Scotland (SQA) · Higher specimen Paper 1 Q19

A 3-mark LED answer: what each mark needs: © Qualifications Scotland (SQA) · marking pattern from the Higher specimen Paper 2 Q12(b) MI

Credits (3 of 3)

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Powered by light: Photo: NASA

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Hinge question: p-n junction devices: © Qualifications Scotland (SQA) · Revised Higher 2014 Q19

Extension: Photo: NASA/JPL-Caltech